

TOD Model Recalibration in China

中国TOD模式再构



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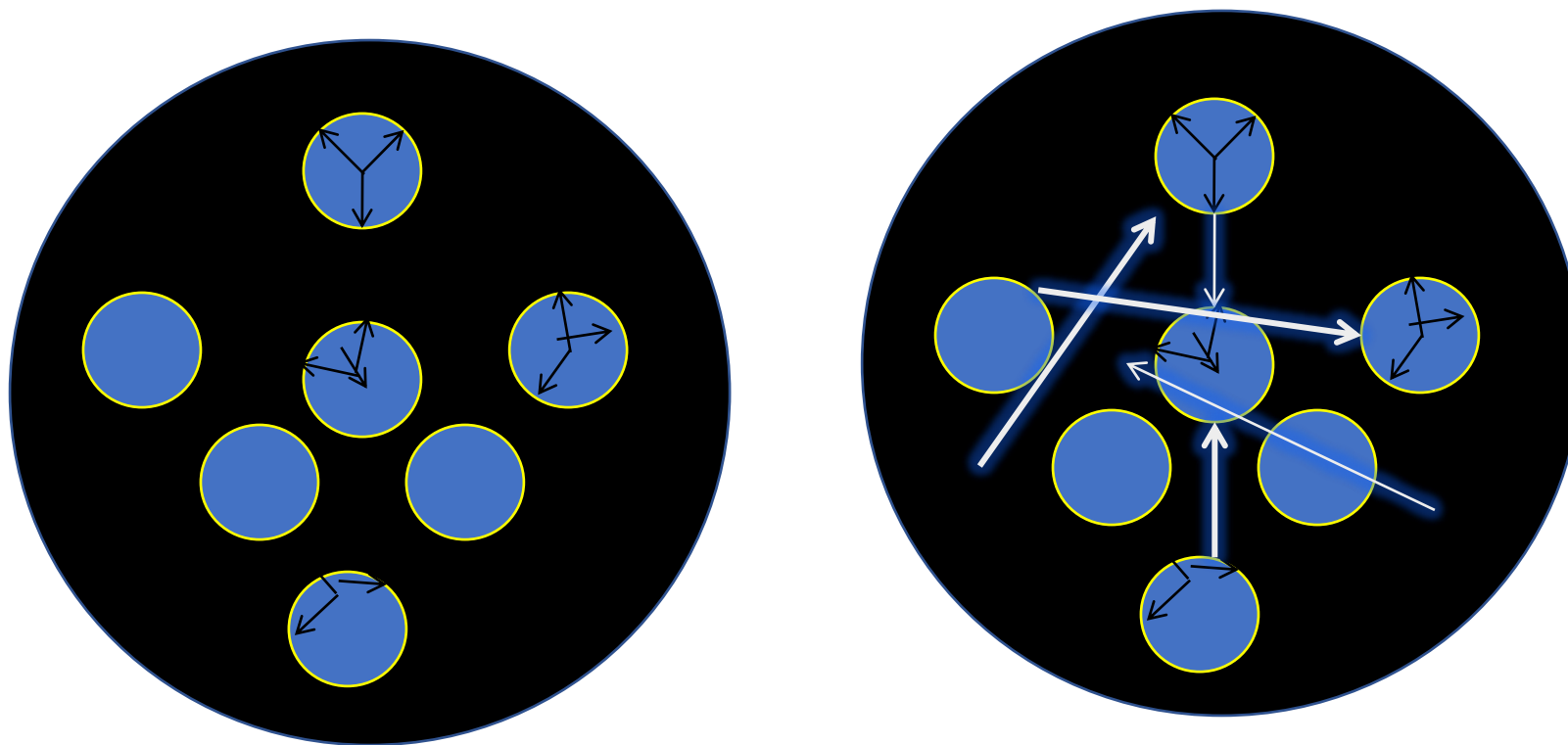
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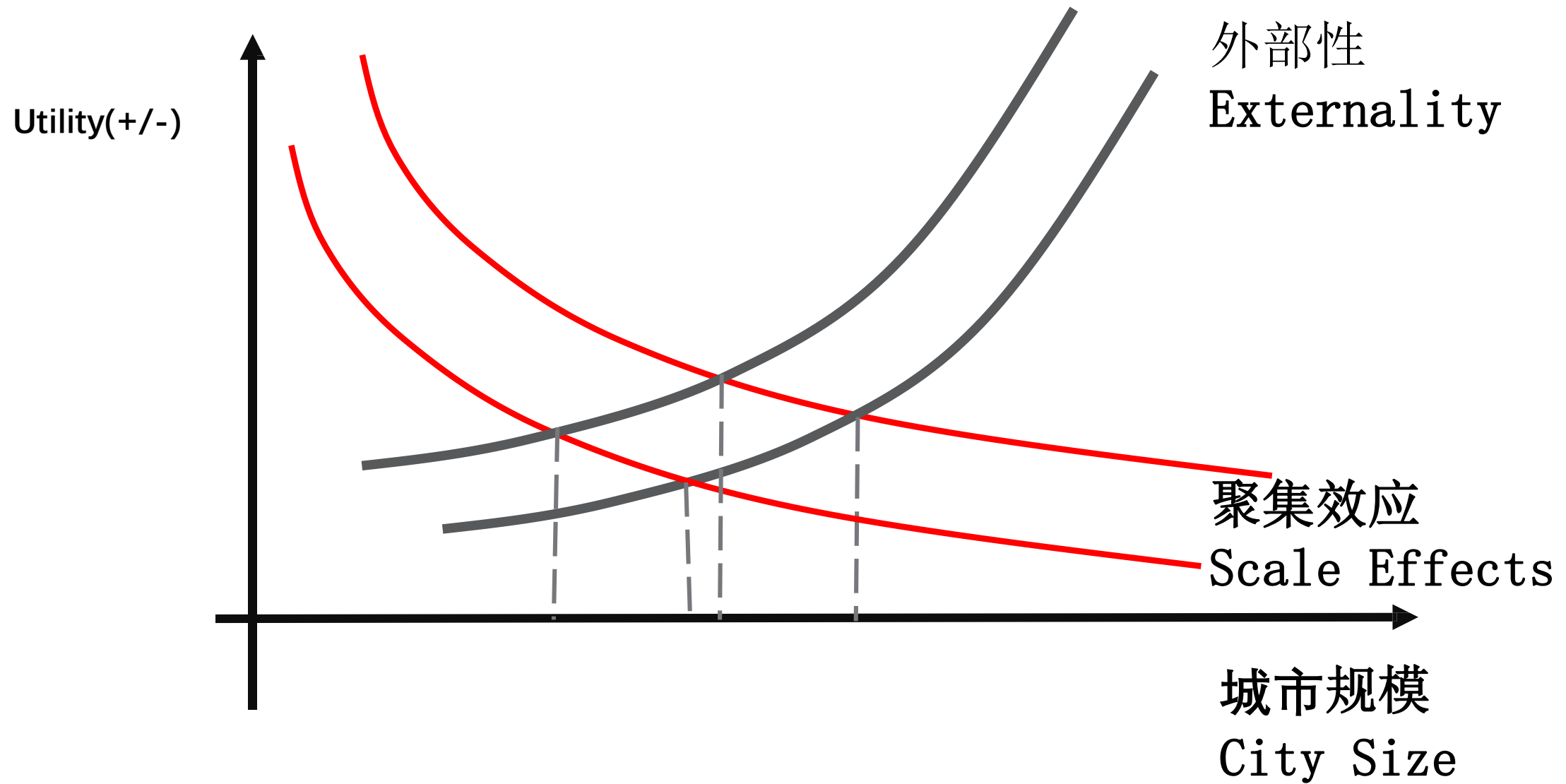
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组团式发展—就业与居住平衡



Less Traffic, but Low City Wide Mobility

政策导向与城市管理 Urban Policy, City Size

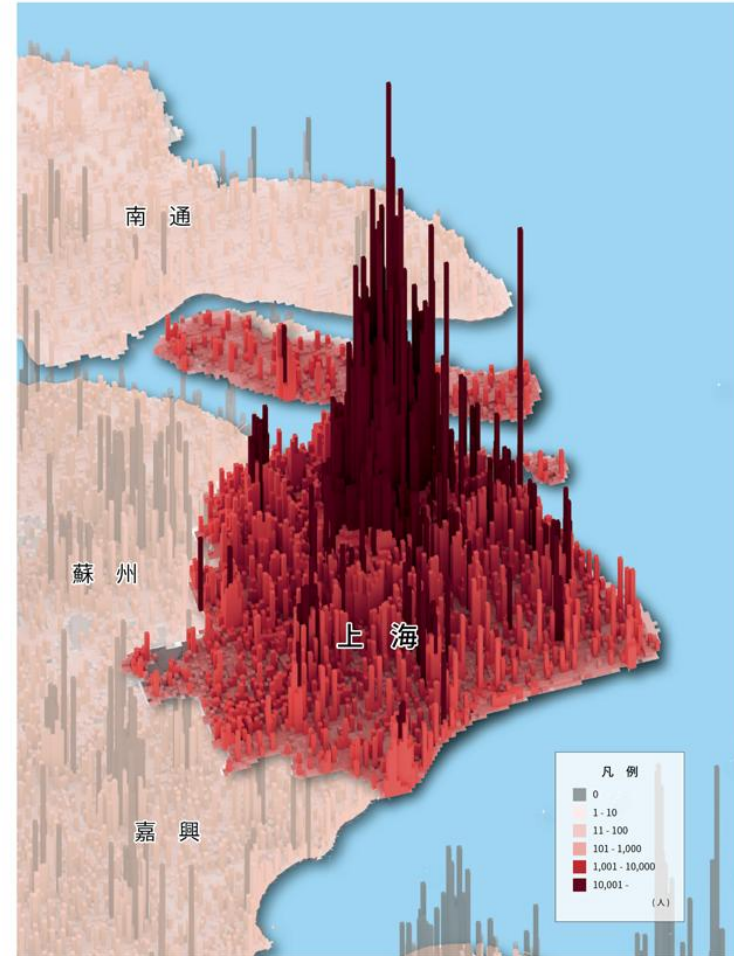


Density was: 200,000 people/sq.km

Living area 4 sq.m/capita



24 Million population in 6500 sq.km
4 million people in 100sq.km



Country of Planning
But urban form does not follow Plan

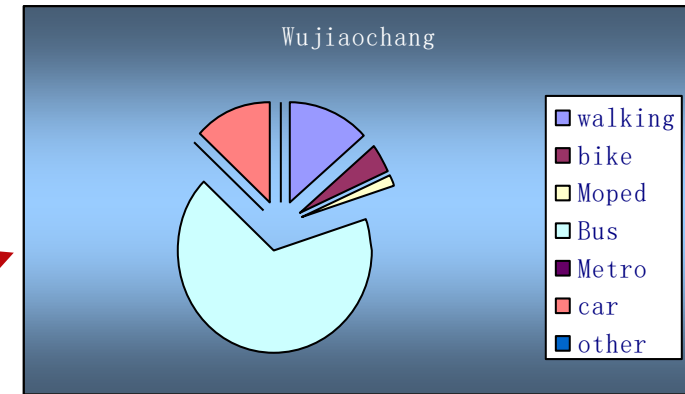
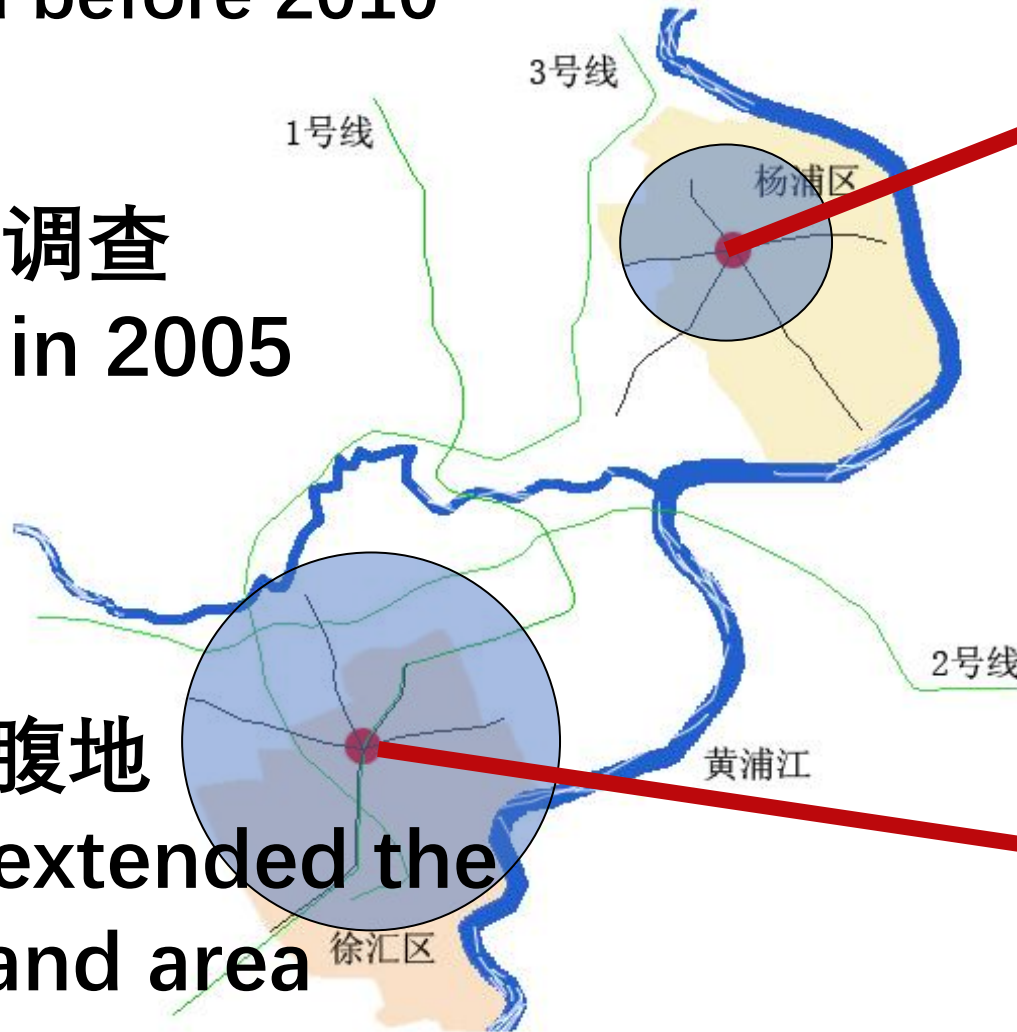
Four Sub-center Planned. Only one
Realized before 2010

2005年调查
Survey in 2005

轨道—

更大的腹地
Metro extended the
hinterland area
Quadruple with Metro

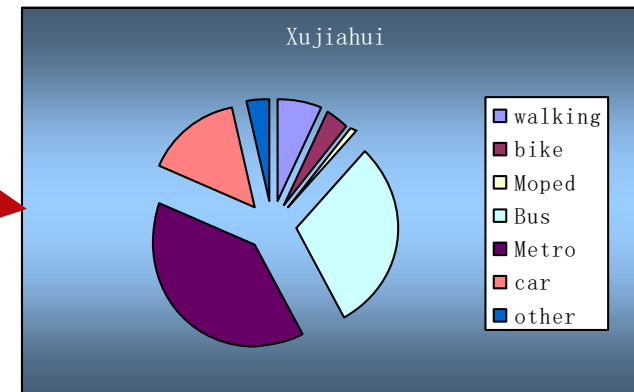
Surveyed in August 2005



Modal Split:

70% by Public Transport

Metro Double the Visitor

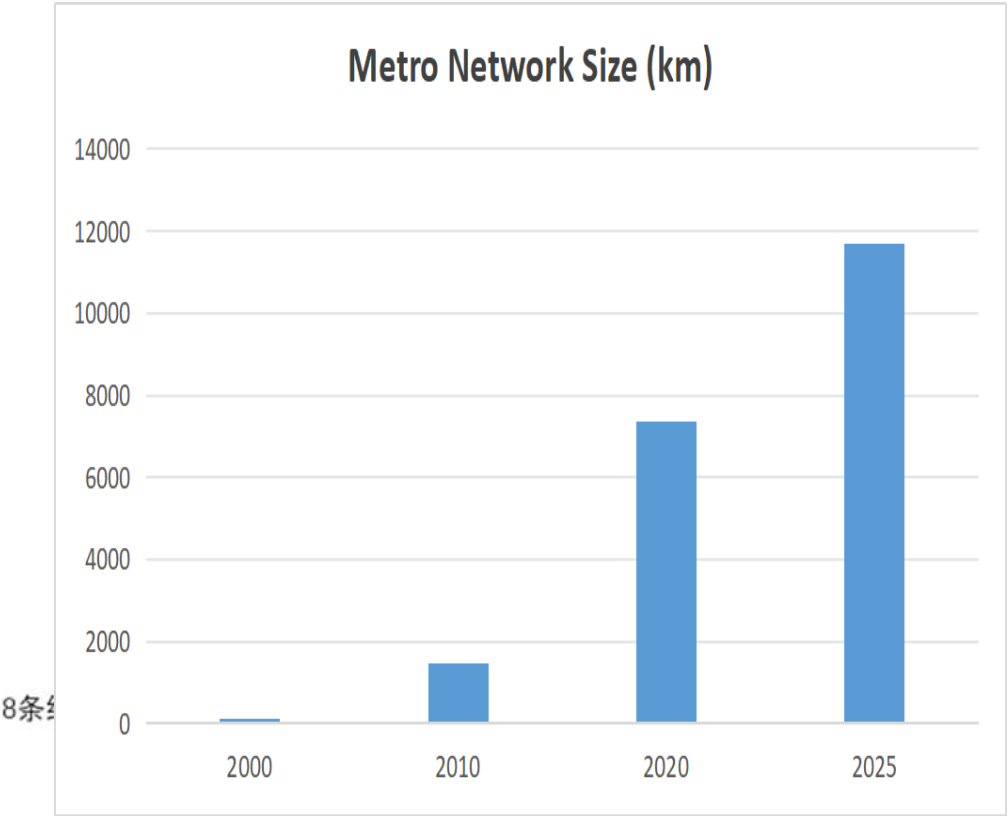
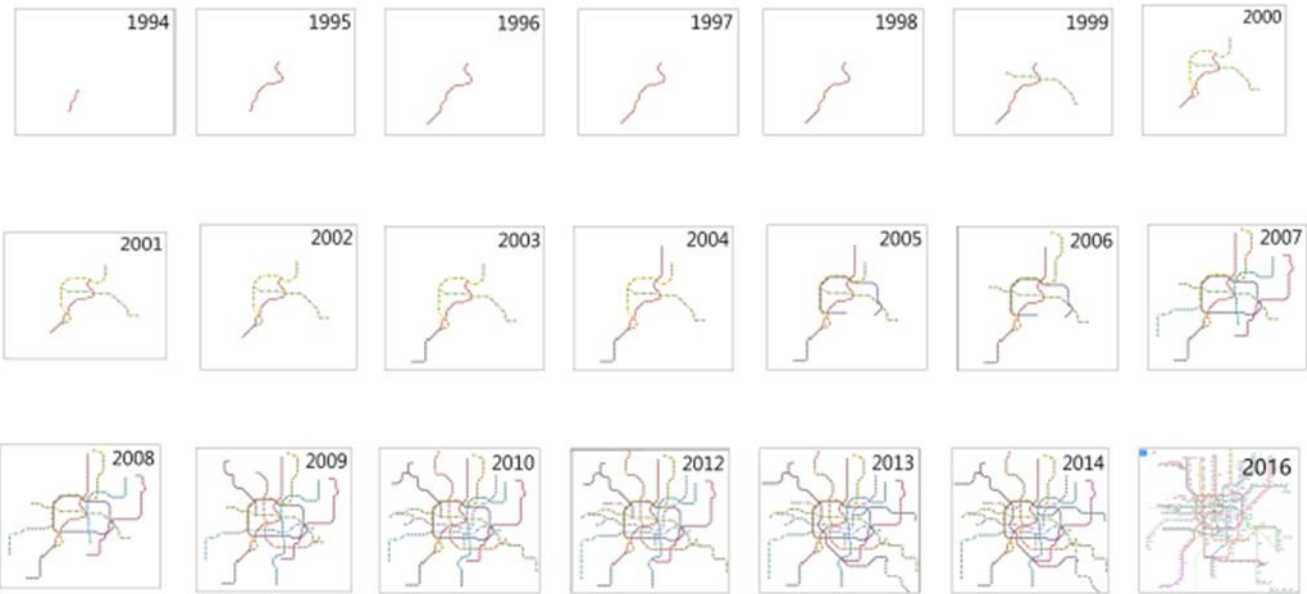


Metro Line 2 Jing-an Temple Station Area, Intensive Commercial Development

Without large scale road capacity expansion



Metro Network in Shanghai



18 Lines

Total Metro Mileage M. China

2004



Speedup construction, very success during 2010 Shanghai WorldExpo

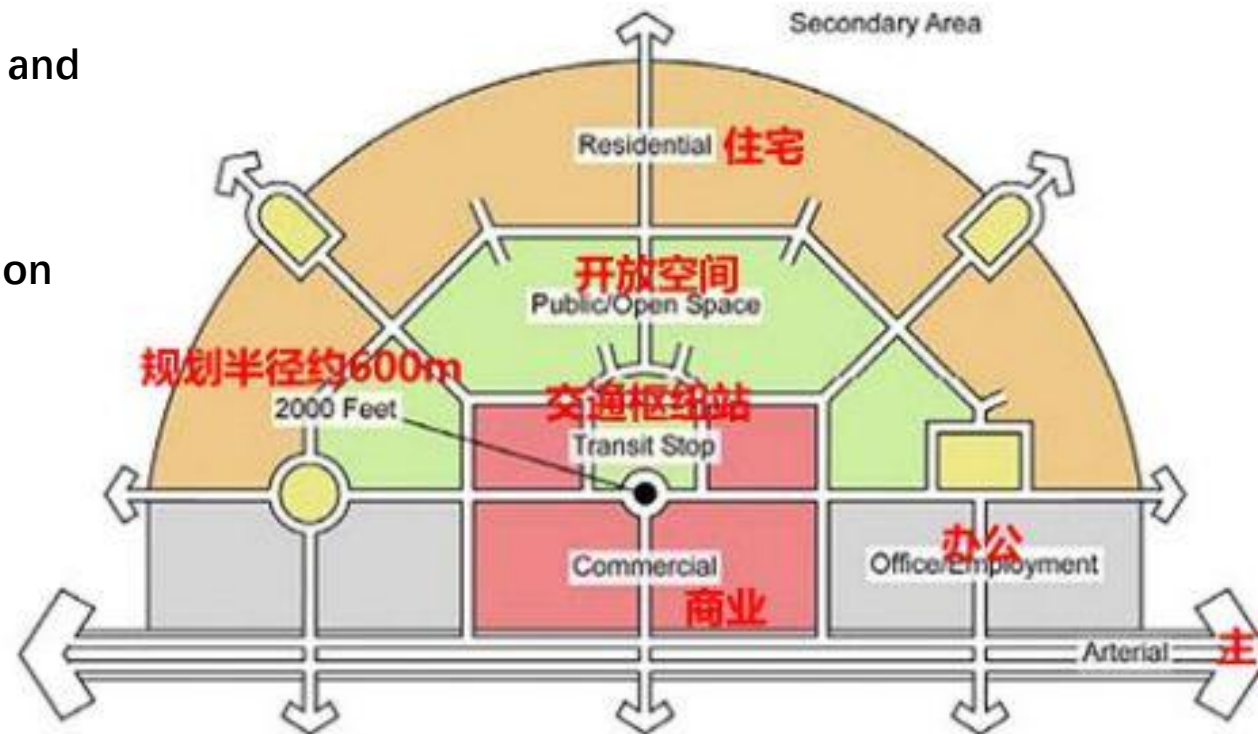
密度、混合、设计3D TOD Model= 设计? Station Area Physical Design?



TOD has been advertised as a cure-all for a host of urban issues, particularly reducing sprawl, decreasing car usage, improving neighborhood access, and creating quality community space (Ibraeva et al., 2020)

TOD must provide ample **pedestrian-scale infrastructure** to **enhance walkability** and other alternative forms of transportation (Ibraeva et al., 2020).

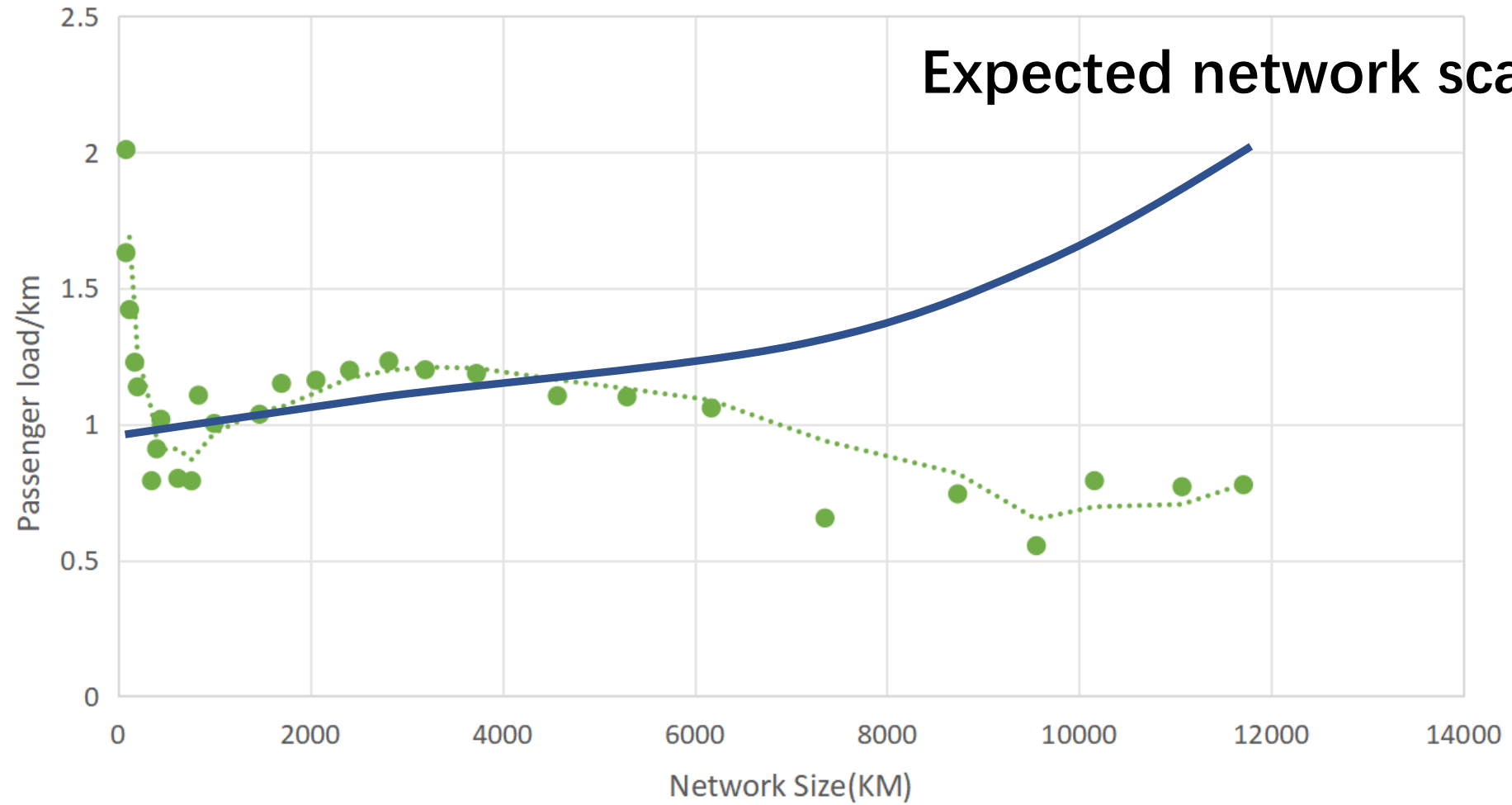
What is the macro condition or **ecology** for a meaningful TOD?



T→D Design? : Huge Attention on Isolated Extravagant Project Architecture Design, Huge Market for Construction Related Business.



Passenger Load/km~Network Size



Expected network scale effect

多模式活动域分析 (Utility 效用)

Multimodal Potential Accessible Space(PAS)

通过一个站点的活动域越大 (多模式网络)，机会越多 (空间特征)，这个站点地区的发展潜力越大，采用轨道的可能性越高

Larger PAS * Larger Activities → High Potential of TOD, design will be more meaningful

多模式网络*空间特征 这里是乘号，不是加号

Multimodal Network * Spatial Characteristics

Not Plus+

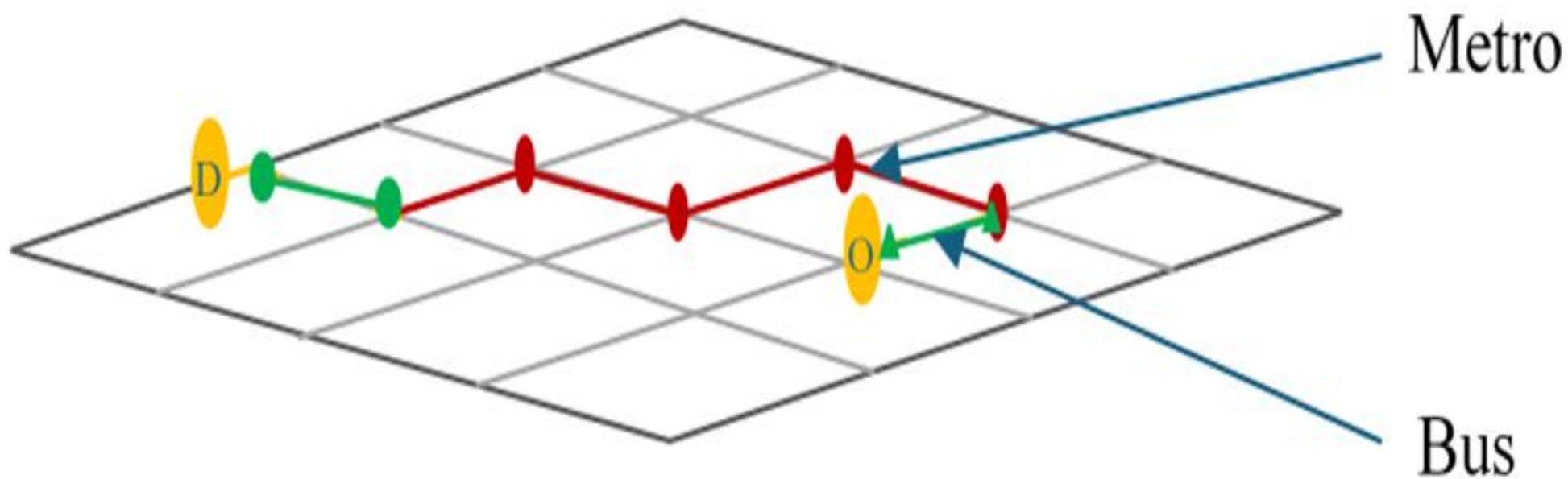
Metro+ Bus, E-Two-Wheels, Bike, Walk

公交+地铁，或电动两轮车加地铁

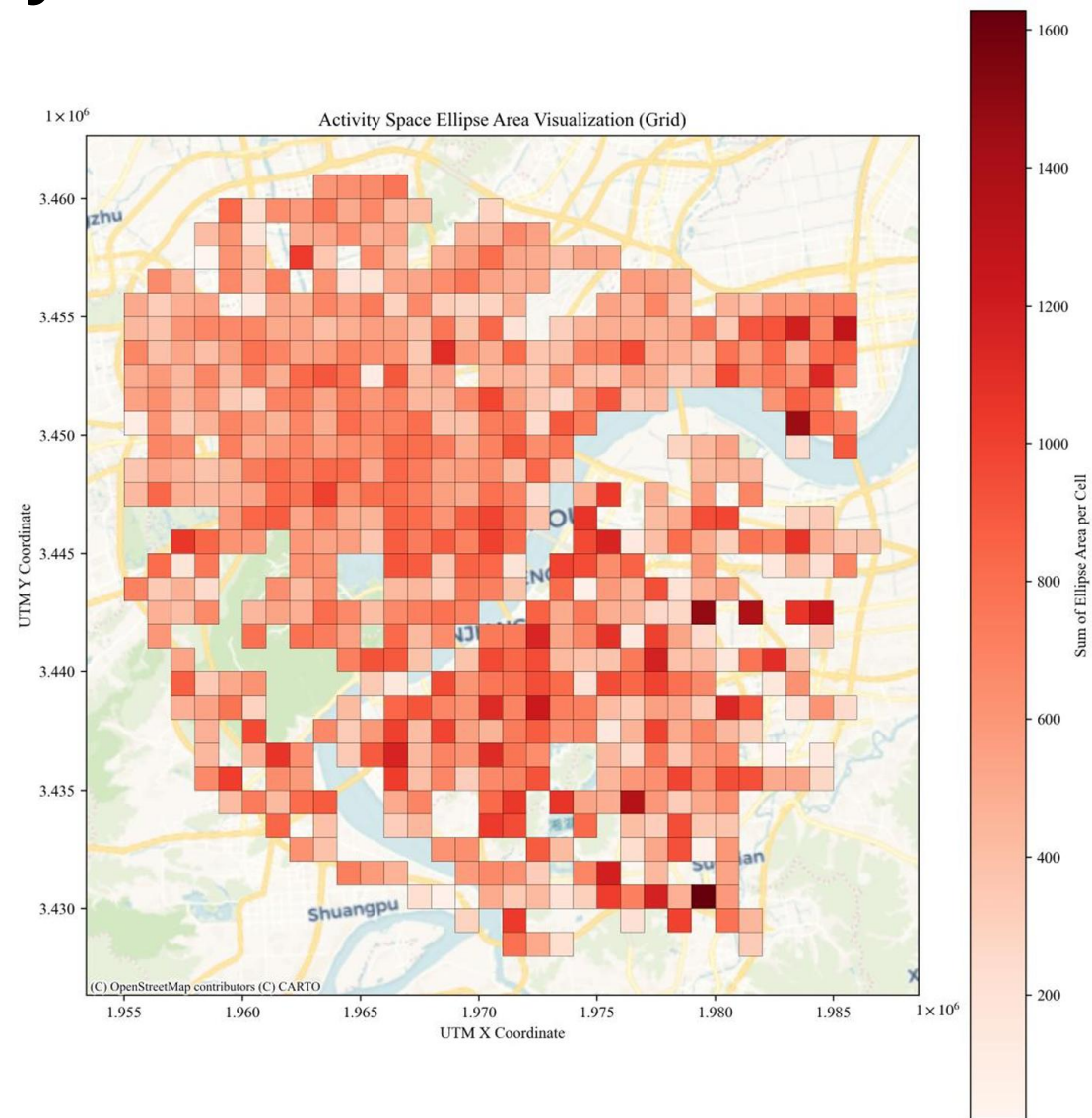
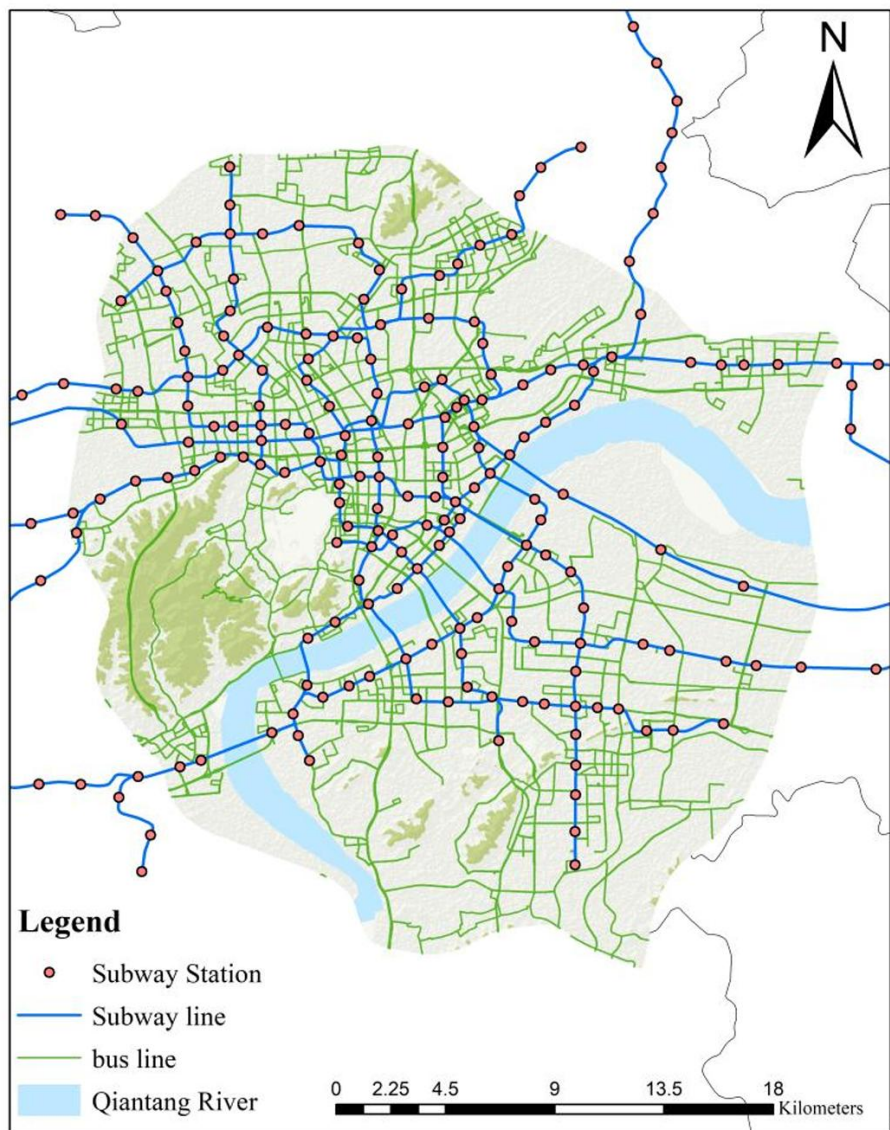


Metro+ Bus, E-Two-Wheels

公交+地铁，或电动两轮车加地铁

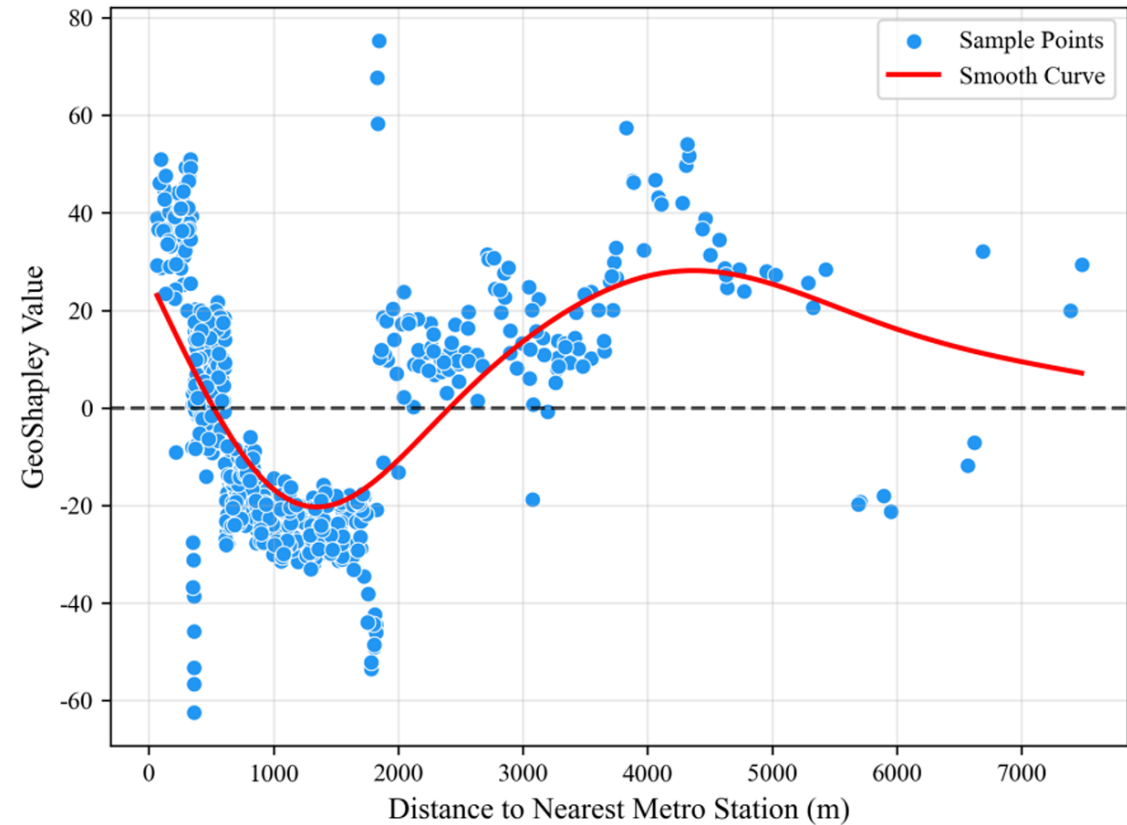
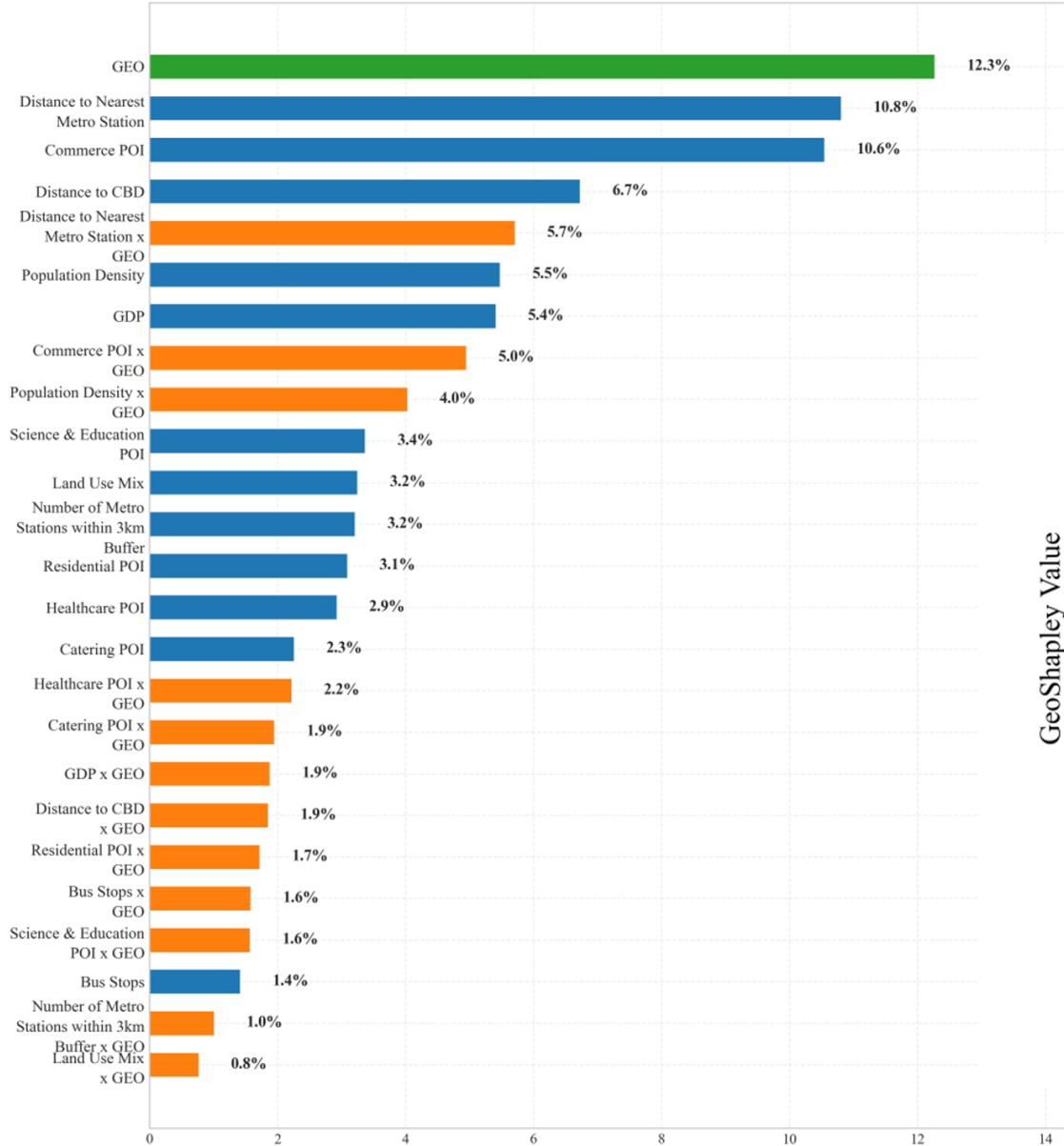


Spatial Heterogeneity 空间的异质性



Impacts of location and built environment elements

GeoShapley Feature Importance Ranking



城市交通政策的影响 Urban Transport Policy

许多城市提出“双快”的城市交通战略，早期道路建设的巨大贡献，这种战略深受决策者的欢迎，体现出“路径依赖”的特征。城市为道路拓展的各种准备非常充分，民众欢迎宽马路的建设。

Many cities have proposed a “Dual-Fast” urban transportation strategy

More Freeway and More Metro

Significant contribution of early road construction project has made this strategy popular with policymakers, reflecting a "path dependence" characteristic.

Cities have made thorough preparations for big road expansion, and the public welcomes the construction of wider roads.

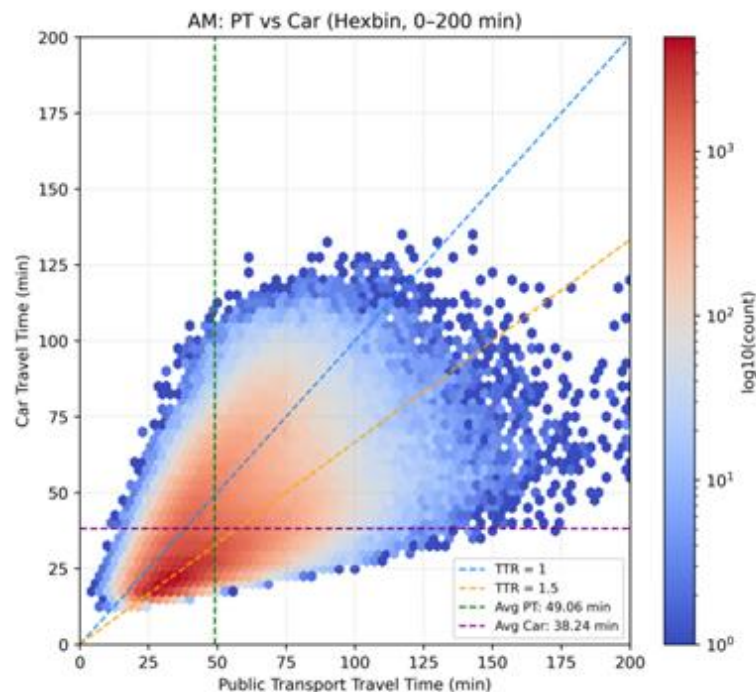
1. 保证儿童上学汽车接送的方便性
2. 大型医院缺乏公共交通外迁
3. 重点建设区域缺乏公交，或公交难以维持

与小汽车相比大规模投资的轨道系统是否还有竞争优势？

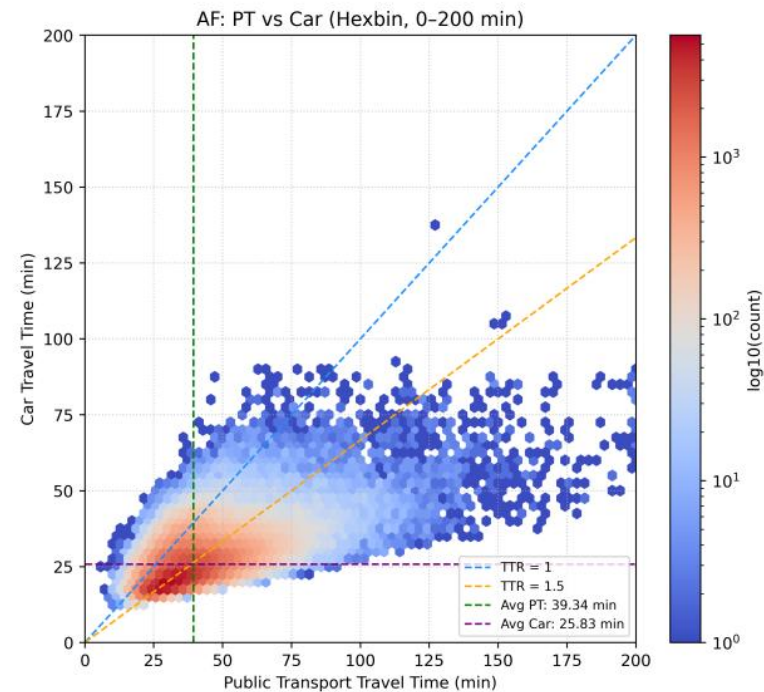
1. Ensuring convenient car transportation for taking children to and from school;
2. More and cheap car parking;
3. Large scale key government development without public transportation or difficult in maintaining the economy of public transportation operation

Does a large-scale rail system still have a competitive edge compared to cars?

高峰时间 Peak



平峰时间 Off Peak

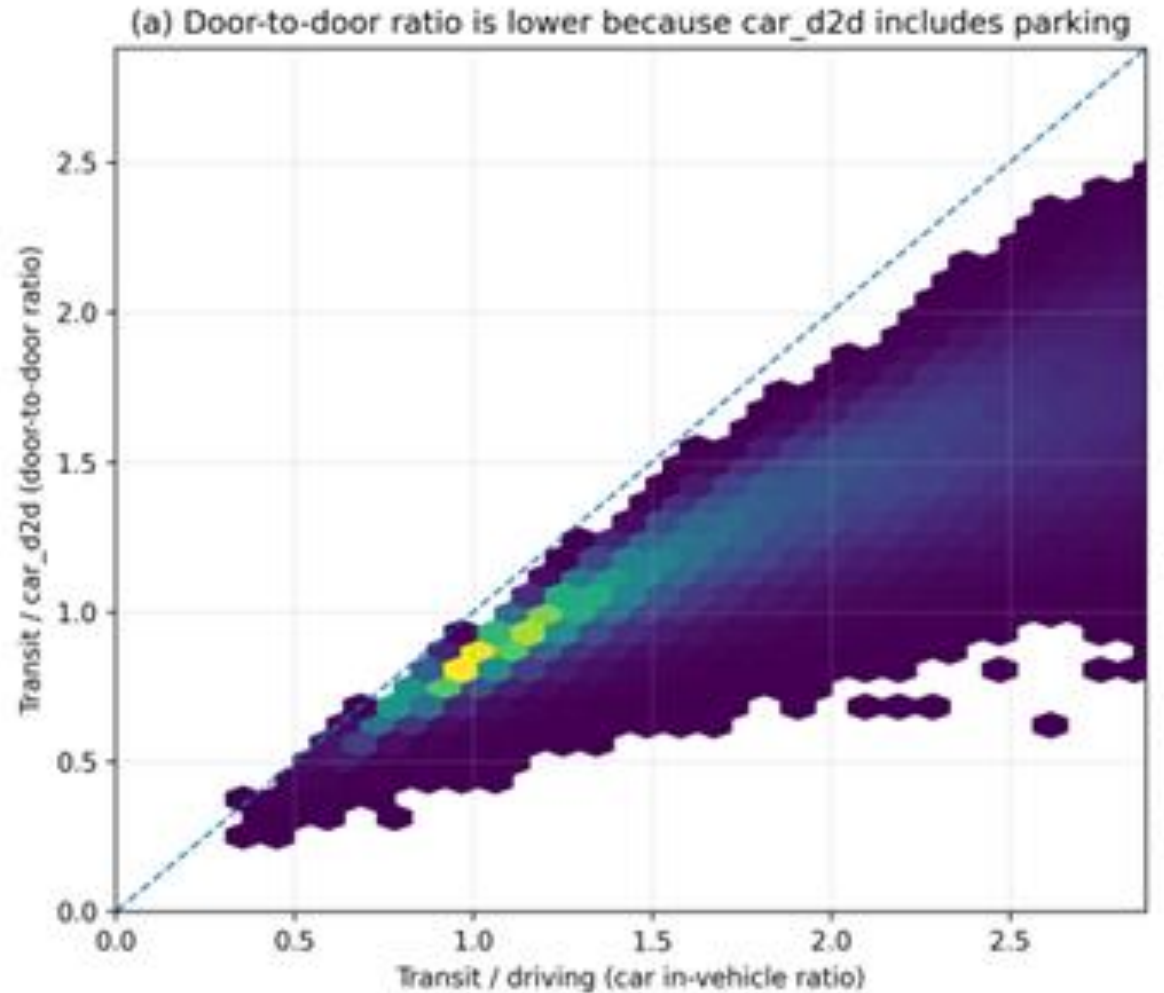


平峰时间基本上没有优势

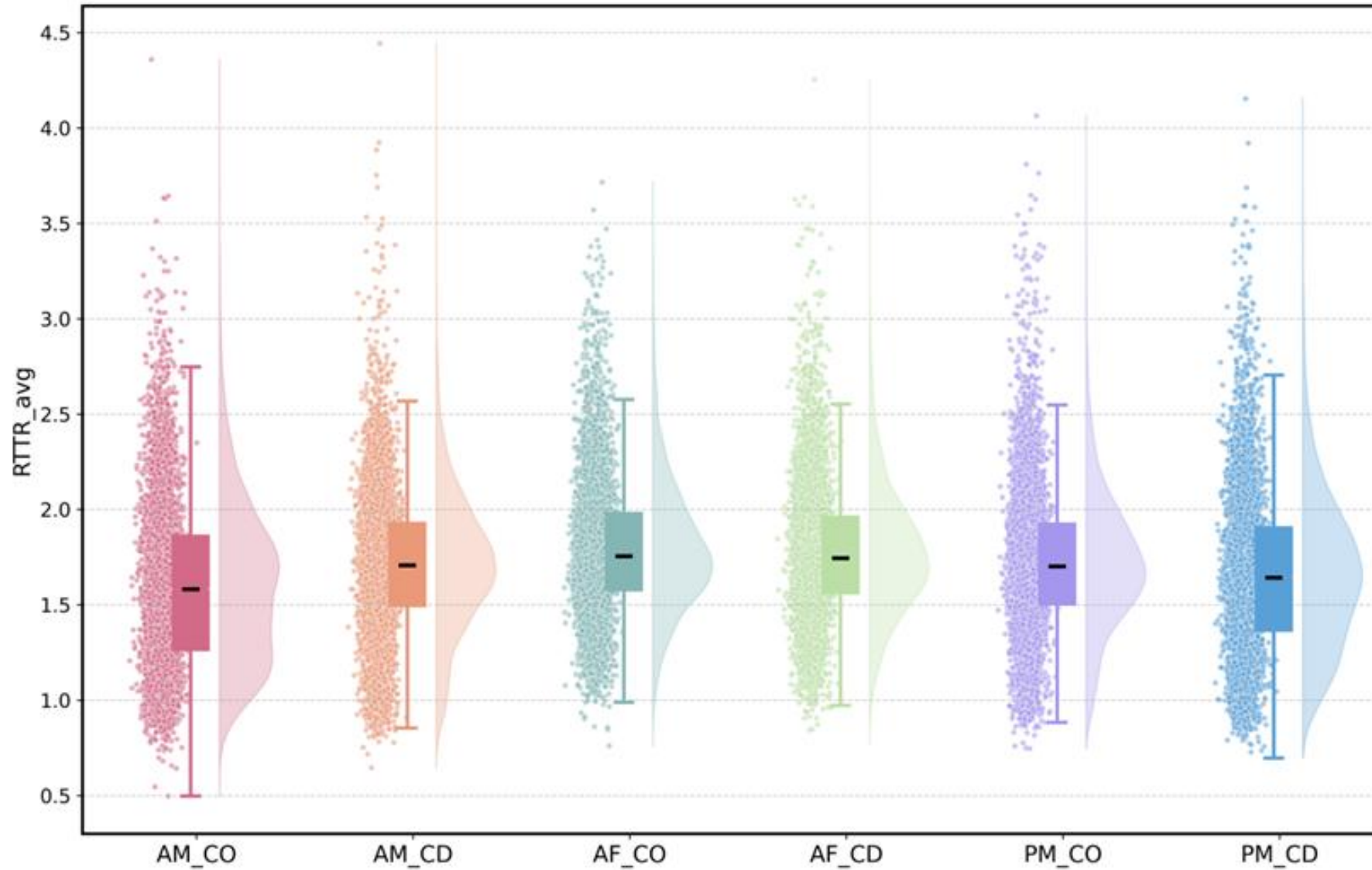
0.28% active territory has advantage over car in off peak

5.05% in peak time

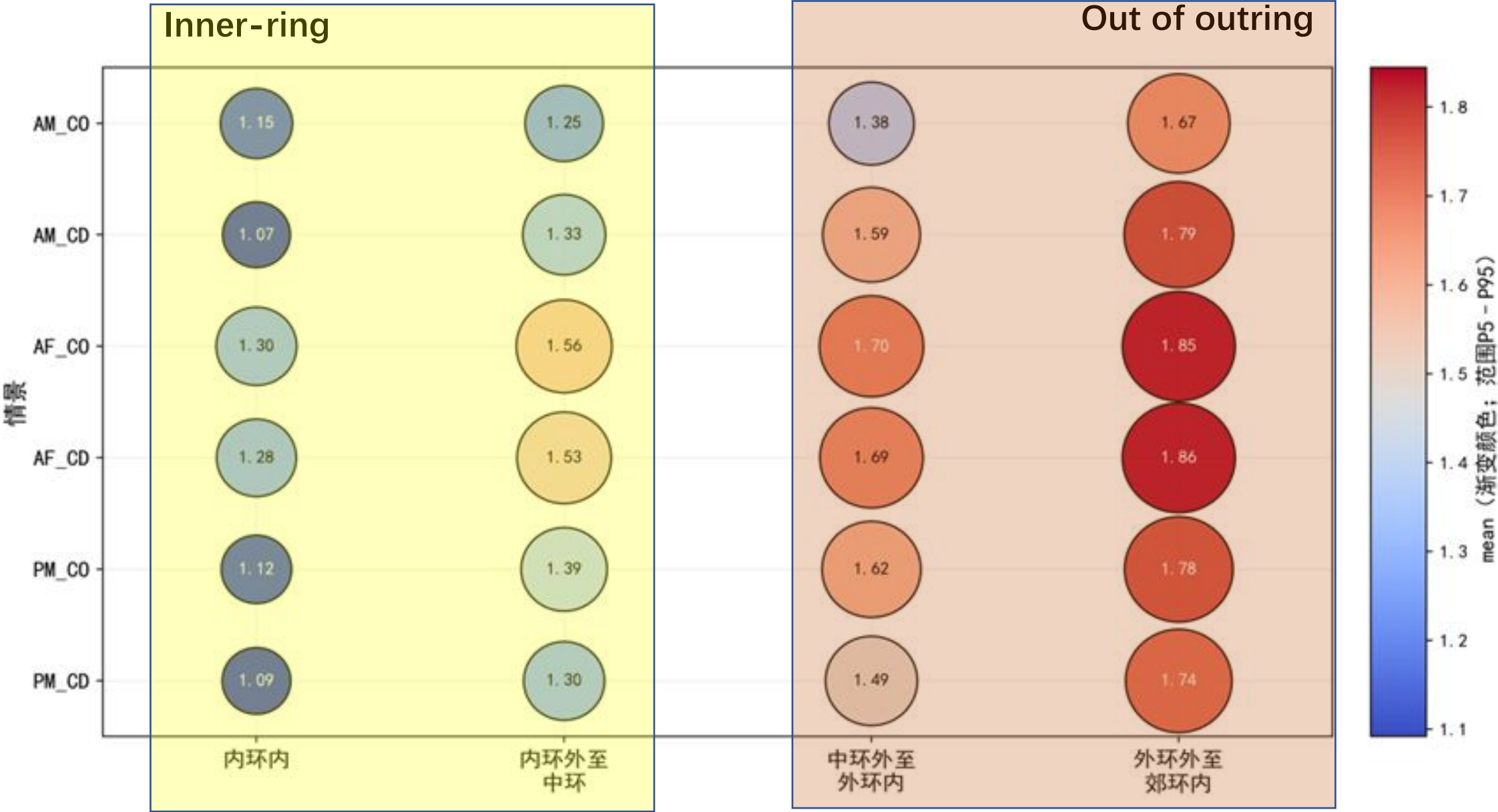
Parking can adjust travel time ratio



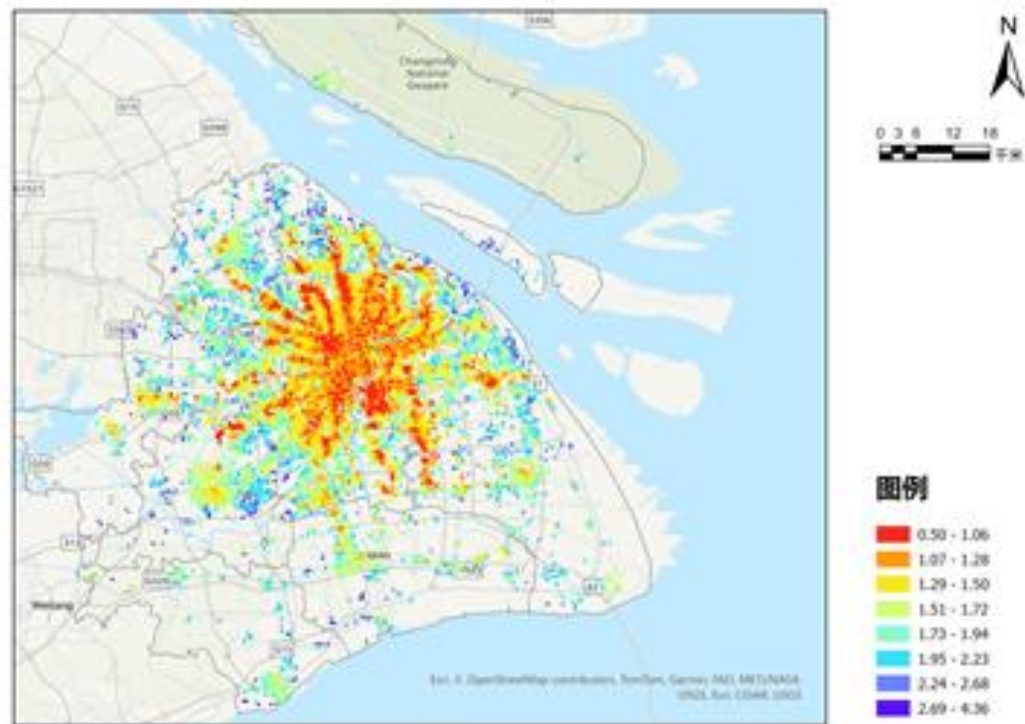
Travel time ratio by origin and destination



Time Ratio by Belt

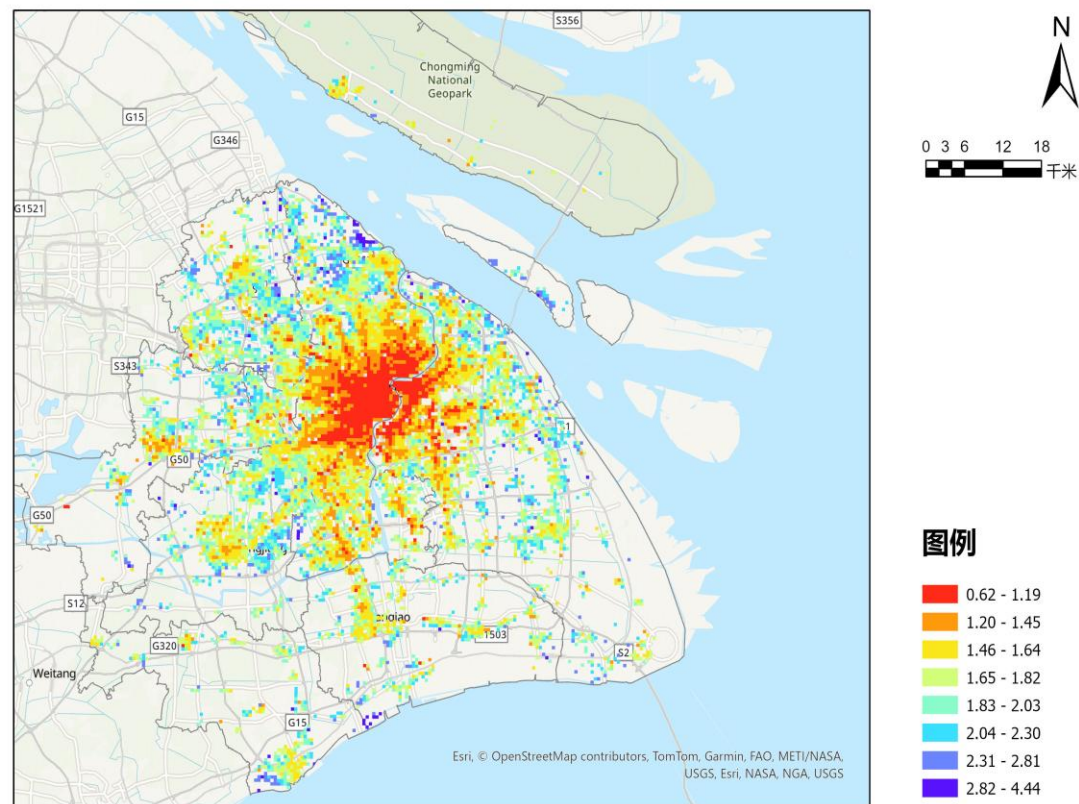


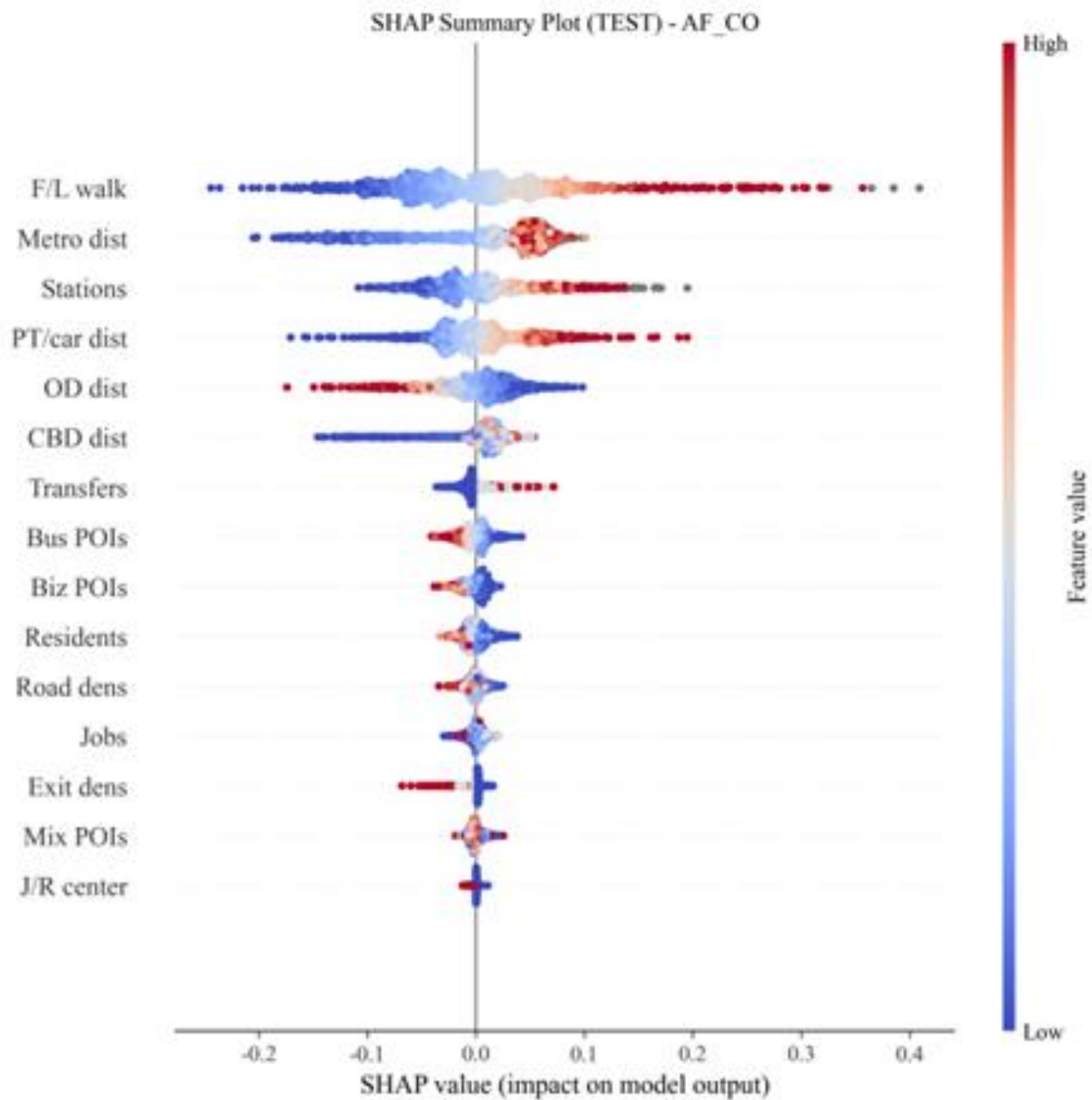
Current de-centralized planning is
encourage car dependence and **Against**
TOD



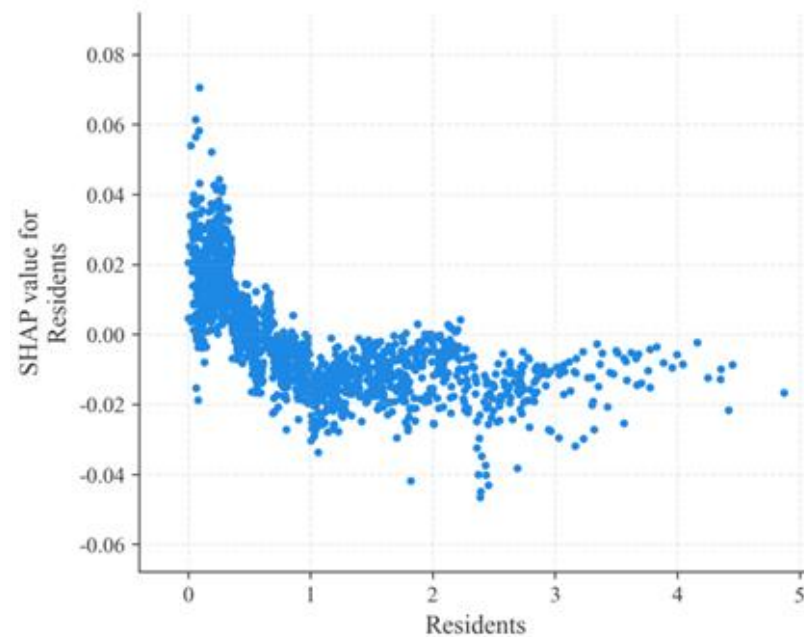
高峰出发端
Origin

高峰目的端
Destination

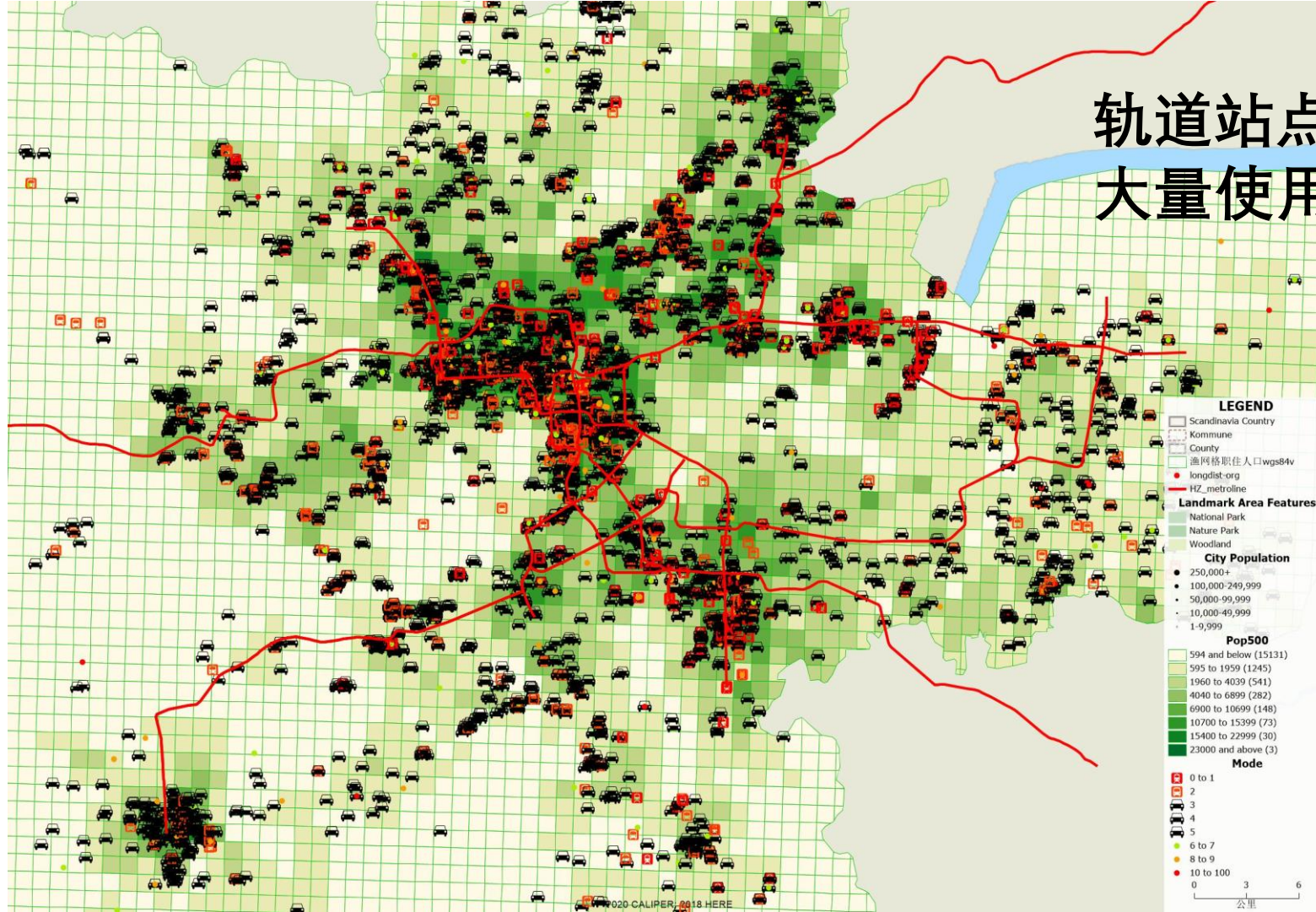




密度的非线性影响



Car is still extensively used for people reside a station area



杭州驾车与轨道交通时间---》客流量

Cartime-Metrotime and Pair Passengers

驾车>乘地铁 时间差前五	驾车-乘地铁 时间/分钟	距离/千米	OD客流量	驾车<乘地铁 时间差前五	驾车-乘地铁 时间/分钟	距离/千米	OD客流量
萧山国际机场站- 向阳路站	17	2.79	385	良渚站-临平站	-75	25.77	261
钱江世纪城站- 潘水站	16	10.49	482	杜甫村站-临平 站	-54	24.37	137
萧山国际机场站- 永盛路站	16	3.72	1487	禹航路站-临平 站	-49	37.82	131
杭州大会展中心站 - 下沙江滨站	16	3.44	169	火车南站-临平 站	-48	25.96	175
学院路站-庆菱路 站	13	6.70	459	良睦路站-临平 站	-47	31.55	176

OD客流量 前五	OD客流 量	距离/千米	驾车-乘地 铁 时间/分钟
火车东站-龙翔桥站	6403	6	1
火车东站-文泽路站	5172	13	-5
火车东站-城站	5141	6	-1
火车东站-武林广场 站	5141	5	0
火车东站-凤起路站	5041	6	-1

Pair Dest.		Time Diff.		Pop.	Emp.
解释变量	截距	站与站之间 距离	驾车与乘地铁时 间差	500米范围 人口数量	500米范围 岗位数量
回归系数	137.387	-4.029	2.239	-0.0000812	0.00606
P值	< 2e-16 ***	< 2e-16 ***	< 2e-16 ***	0.783	< 2e-16 ***

站点地区交通方式选择---嵌套离散模型

Station Area Travel Modal Choice

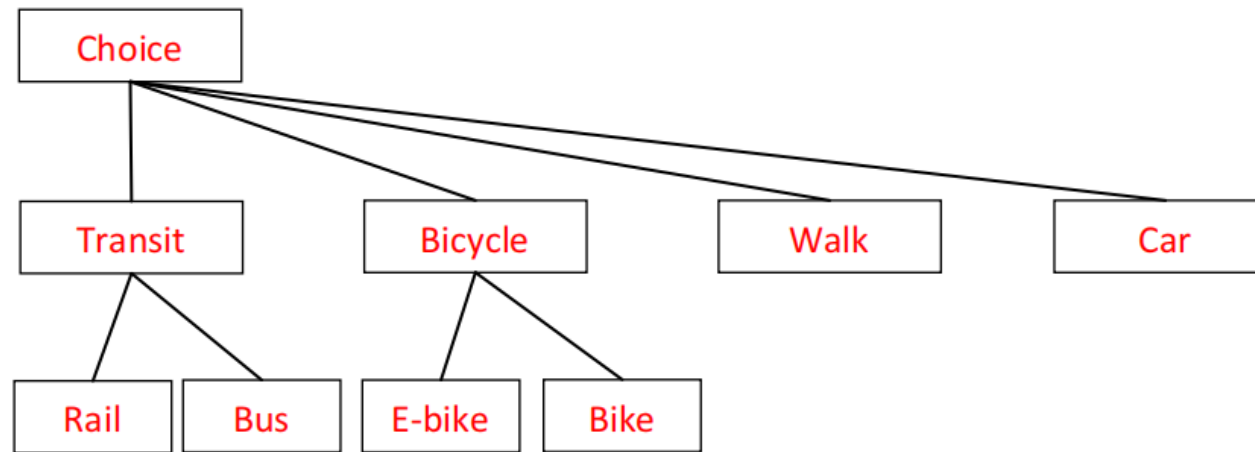


Fig. 3. Structure of the nested logit model.

The choice probability for alternative i for person n , is shown in Eq. (2) (Train, 2003):

$$P_{ni} = \frac{e^{V_{ni}/\lambda_k} \left(\sum_{j \in B_k} e^{V_{nj}/\lambda_k} \right)^{\lambda_k - 1}}{\sum_{l=1}^K \left(\sum_{j \in B_l} e^{V_{nj}/\lambda_l} \right)^{\lambda_l}}$$

大量城市范围的规划要素

Urban planning elements

选择轨道交通的影响因素, **Impact factors of metro riding**

自变量	B	S. E.	Wals	Sig.	Exp (B)
助动车**	-1.4887134	0.336	19.638	0.000	0.226
自行车	-0.3463871	0.246	1.985	0.159	0.707
小汽车拥有情况**	-2.0911310	2.067	2.237	0.035	0.124
小汽车牌照(1)	1.72880451	1.608	1.155	0.282	5.634
居住地的停车费用	-0.0216699	0.01	5.079	0.124	0.979
上班地点停车费用	-0.0010042	15.557	0.000	1.000	0.999
上班地点寻找车位时间	-7.5882137	1684.748	0.000	0.996	0.001
上班地点停车后步行距离	-0.0214315	55.026	0.000	1.000	0.979
性别(1)	0.08730067	0.306	0.081	0.776	1.091
年龄**	-0.0264524	0.012	5.24	0.022	0.974
职业(1)	0.41350184	0.896	0.213	0.645	1.512
家庭收入	-0.0000008	0	0.083	0.773	1.000
轨道交通起点接驳时间TIME _{O_to_Station} **	-0.0012298	0.001	6.016	0.014	0.999
轨道交通终点接驳时间TIME _{Station_to_D} **	-0.0023523	0.000	39.877	0.000	0.998
轨道交通车内时间TIME _{车内} *	-0.0000431	0.001	0.005	0.099	1.000
小汽车出行时间TIME _{car} **	0.00254134	0.001	23.357	0.000	1.003
常量	-0.6105536	1.810	0.114	0.736	0.543

Age

小汽车拥有 Car

电动两轮车拥有 ETW

两端接驳时间比在车内
时间重要100倍

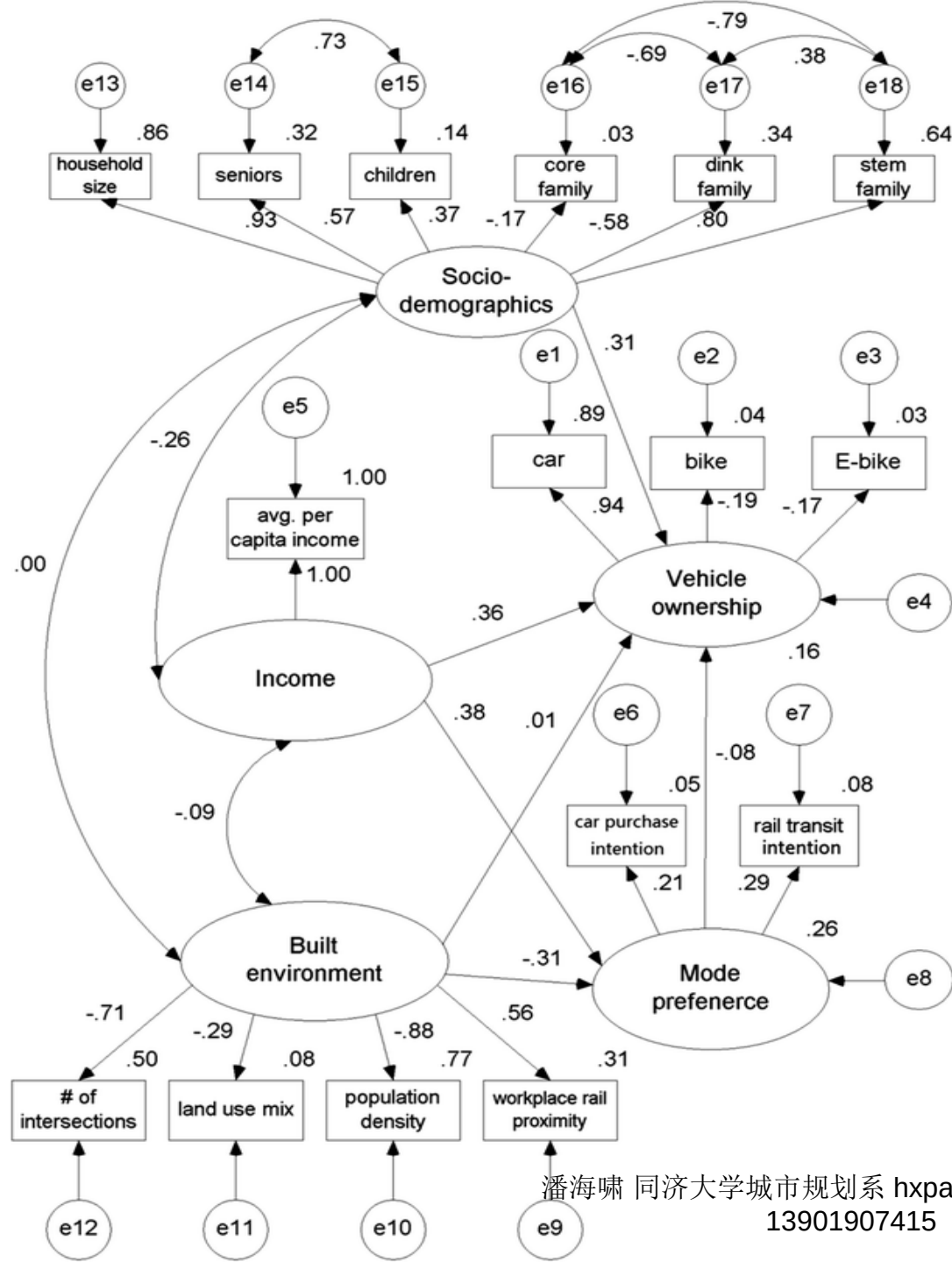
Connecting Time

目的地停车不方便,
寻找车位困难

Parking

到一个目的地开车比
乘轨道时间更长

Short time by metro

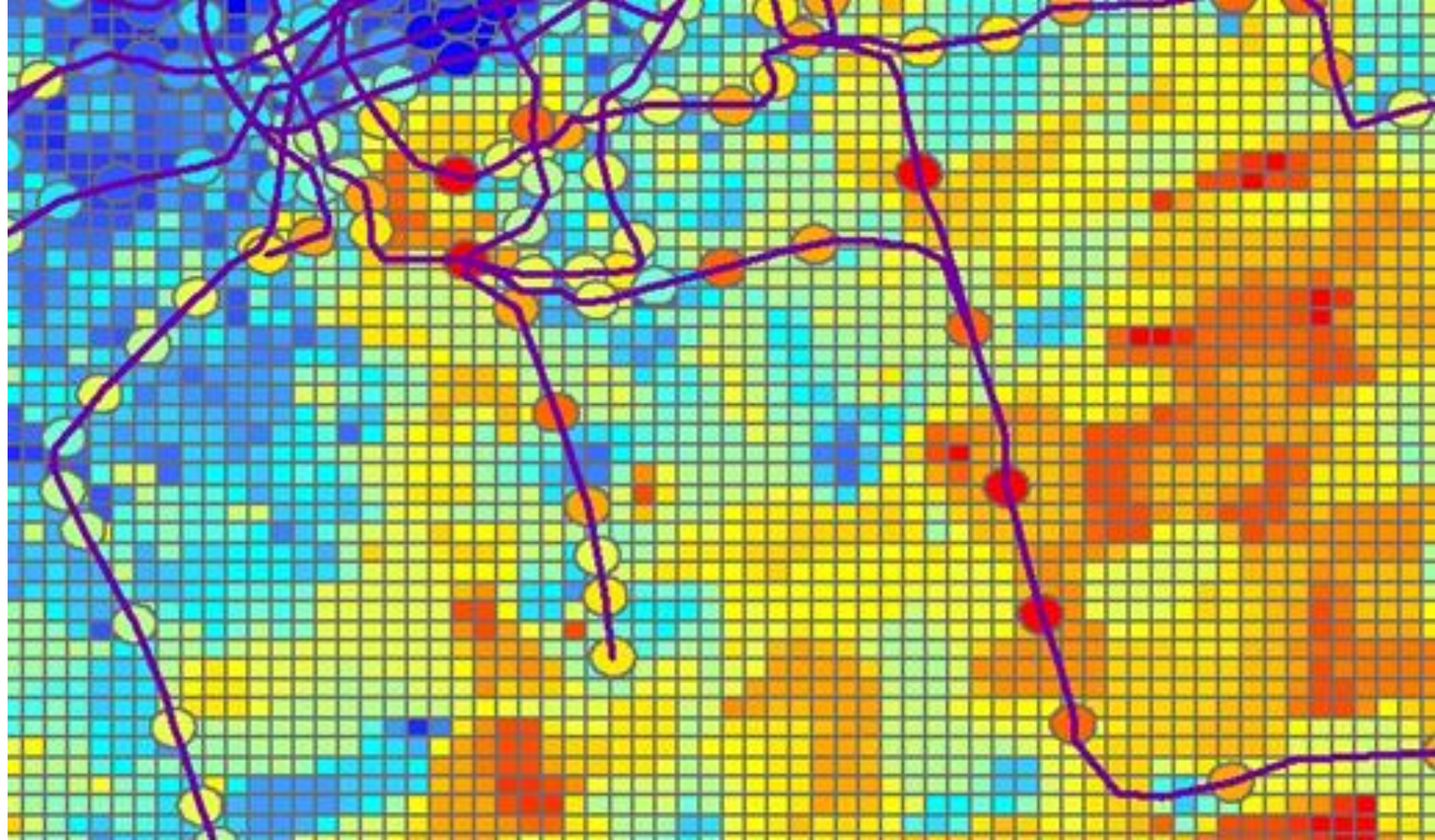


社会经济要素 Social-demographics

收入增加,
Income

有老人 old
Parents

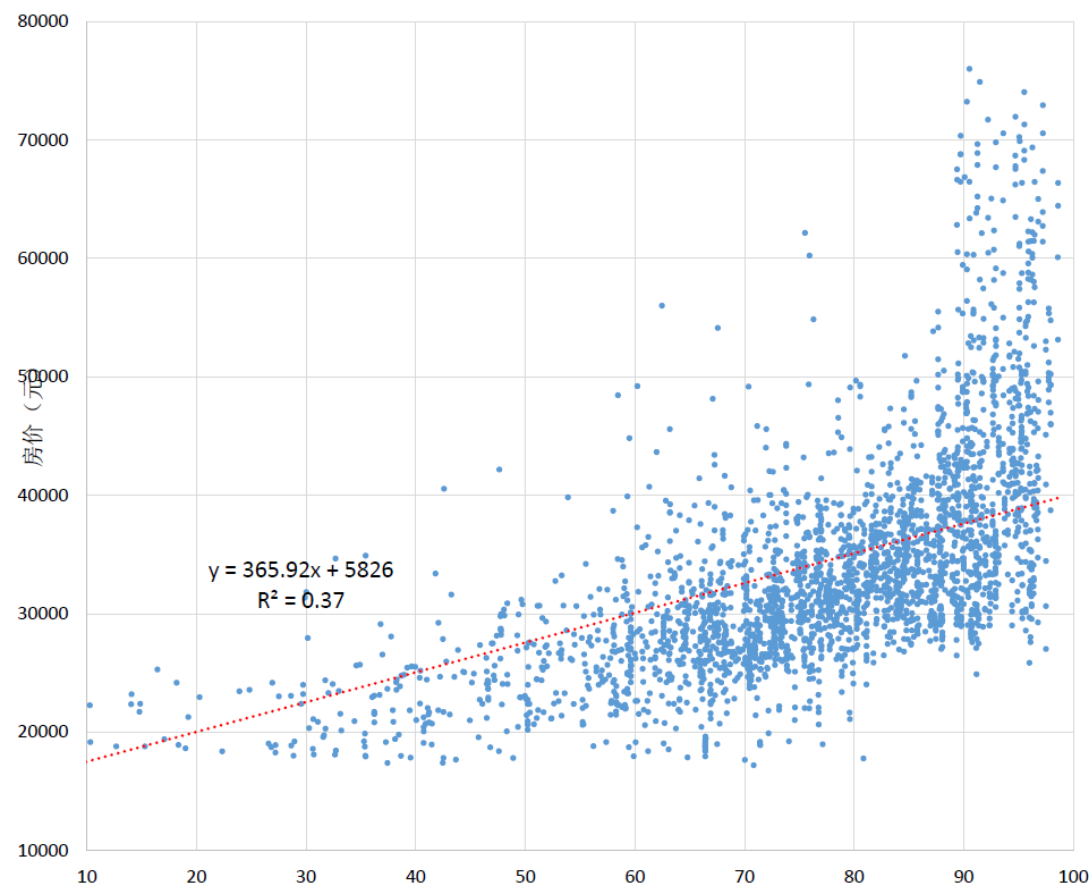
有儿童 Child



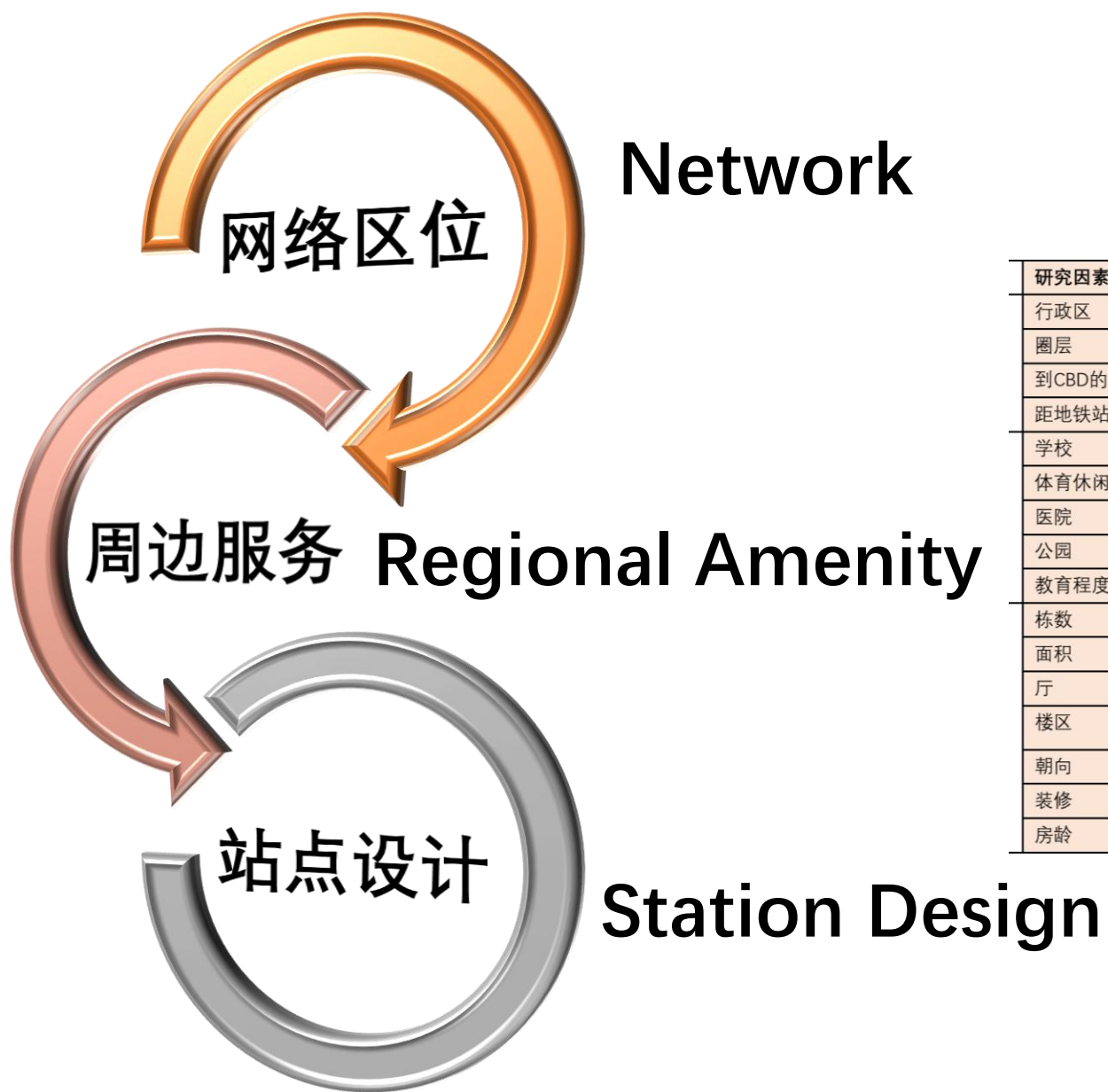
可达性对地价的解释

Network Accessibility and Property price

价格是需求的反映， 没有需求就没有乘客

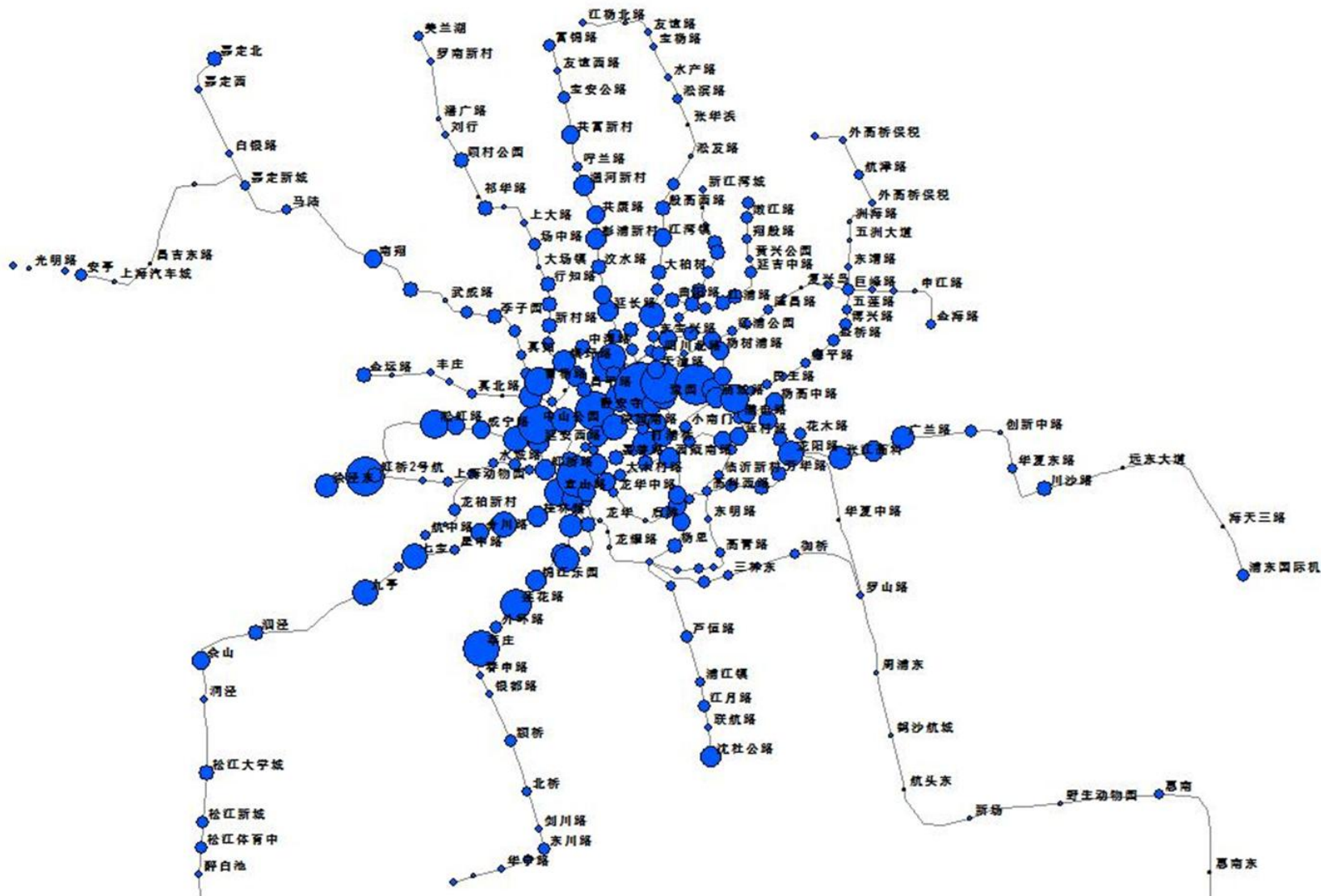


37%的解释率

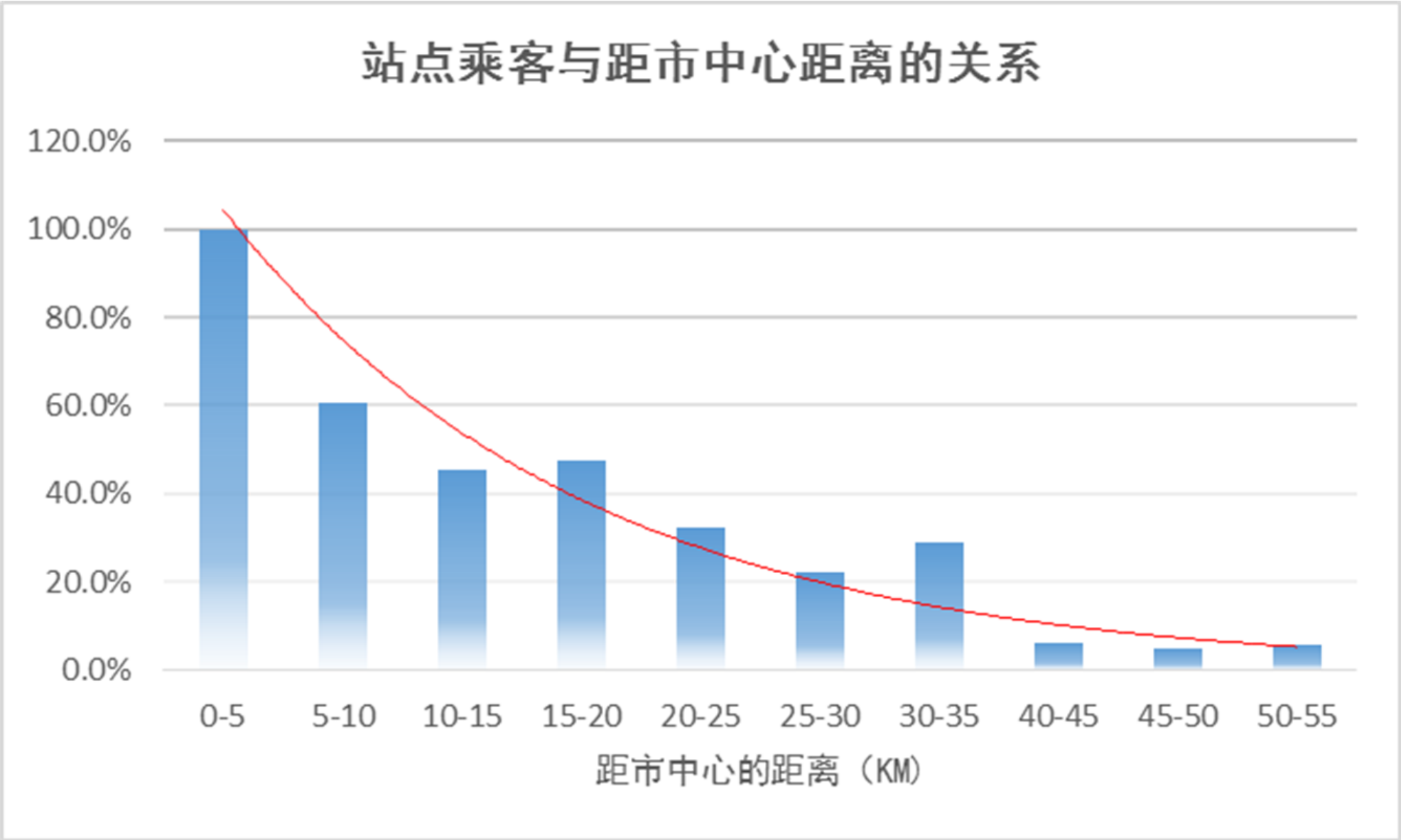


研究因素	单位	各因素水平及说明
行政区		16个区，哑元变量
圈层		4个圈层，哑元变量
到CBD的距离	m	5个CBD，连续变量
距地铁站的距离	m	连续变量
学校	个	500M范围内高校、职校、中学、小学、幼儿园个数，连续变量
体育休闲	个	500M范围内体育休闲设施个数，连续变量
医院	个	500M范围内医院个数，连续变量
公园	个	500M范围内公园个数，连续变量
教育程度		大学+研究生/常住人口，居委会，连续变量
栋数		小区栋数
面积		面积，连续变量
厅		个数，连续变量
楼区		低区、中区、高区，哑元变量
朝向		尚未加入
装修		土坯，简装，中装，精装+豪装，哑元变量
房龄		连续变量

站点客流分析 Station Passenger:



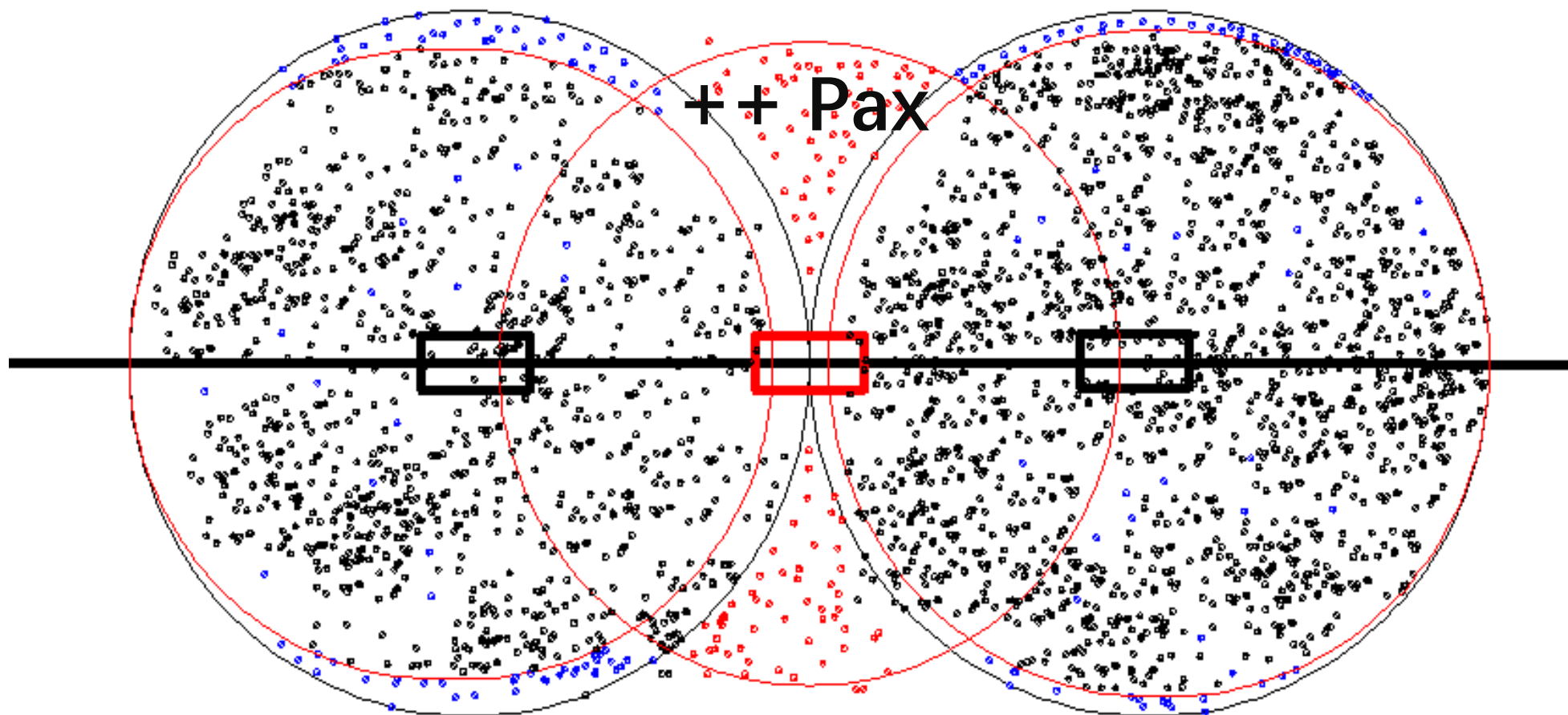
Station Passenger with distance away from city center



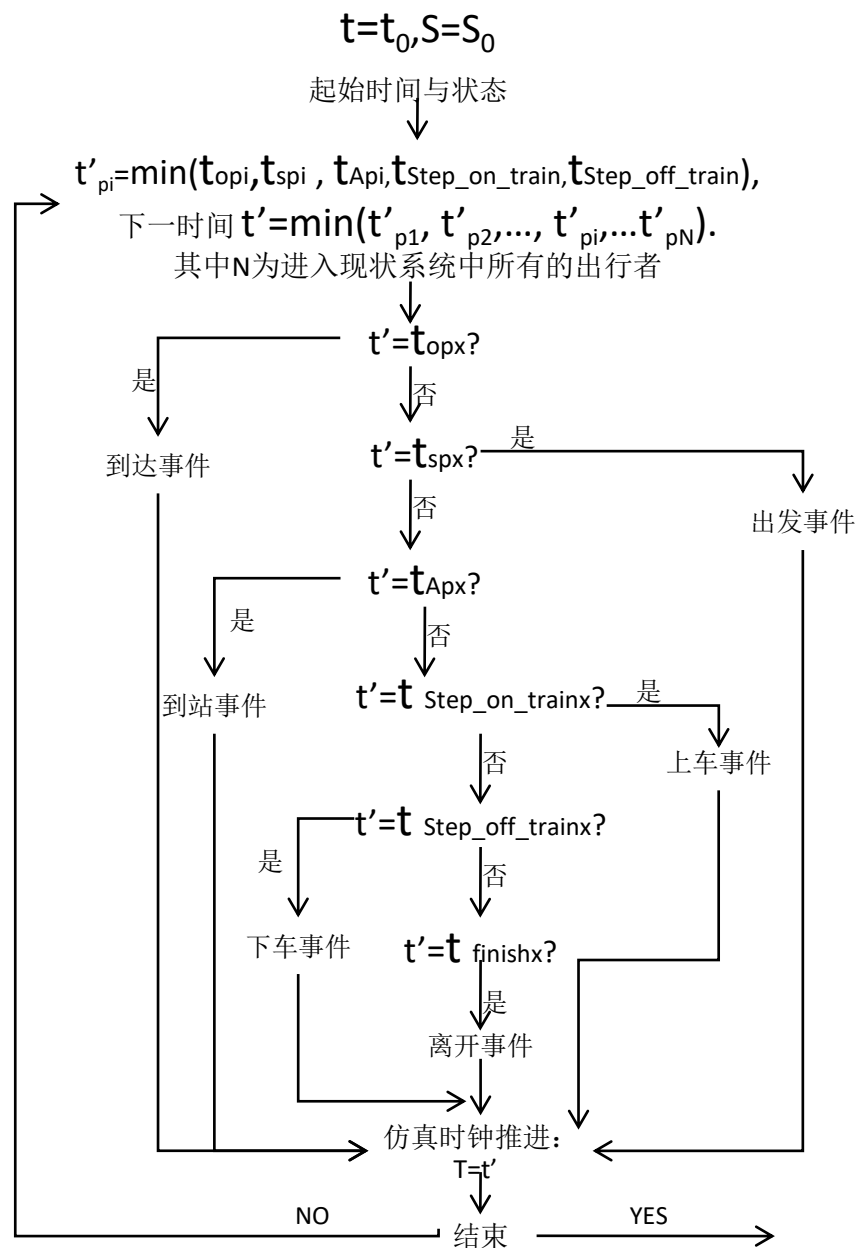
在低可达性地区—郊区， 规划要素 Very difficult to predict, very sensitive
Low robustness

Dependent (Daily passengers)	Coefficients	Standardized Coefficients	t	Sig.
Constant	-6037.05	NA	-1.019	0.310
Employment in 500 m	21.40	0.29	2.916	0.004
Residents in 500 m	-2.51	-0.07	-0.170	0.865
Years since opening	2373.48	0.30	3.689	0.000
Commuting intensity in 500 m	0.44	0.10	0.276	0.783
Two-line transfer	27020.01	0.17	2.115	0.037
District center	21685.93	0.24	3.047	0.003
model	sig. 0.000		adjusted R ²	0.302

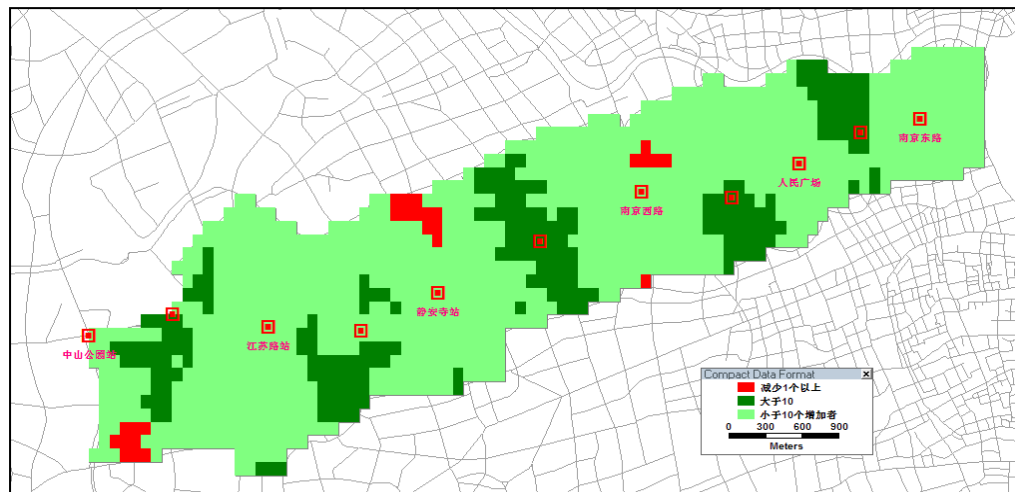
站点设置 Station space simulation - Pax



仿真流程图

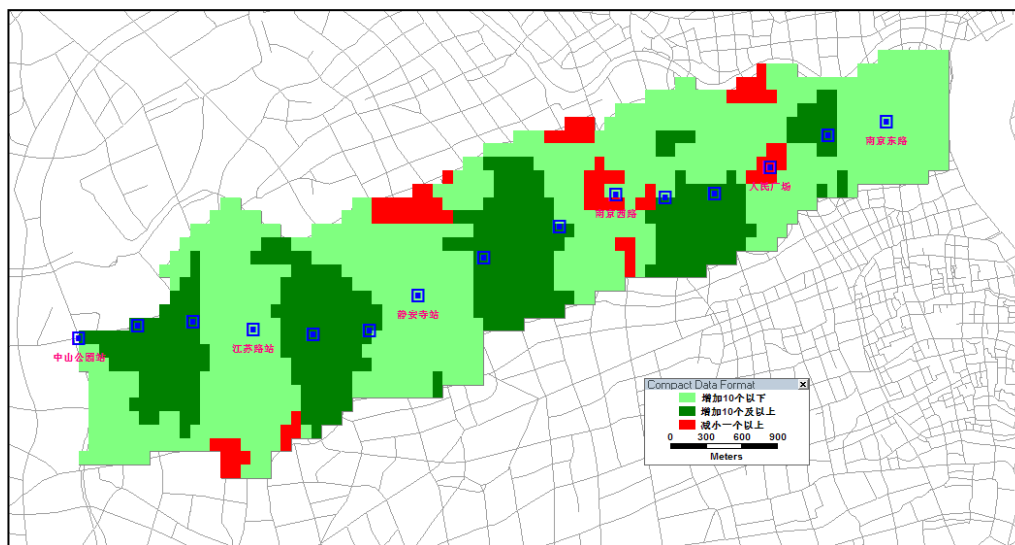


Passenger Increasing



从站距1500米到750米的影响范围变化

✓ 站距减小到700米，轨道交通影响范围变大。



从站距1500米到500米的影响范围变化

✓ 站距减小到500米，轨道交通影响范围变大。

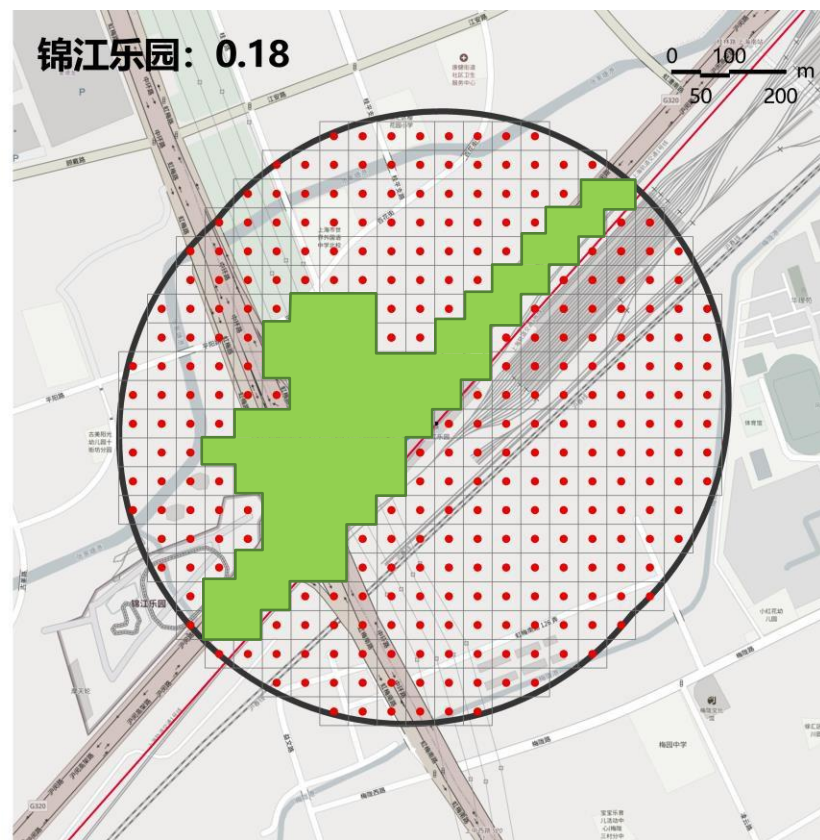
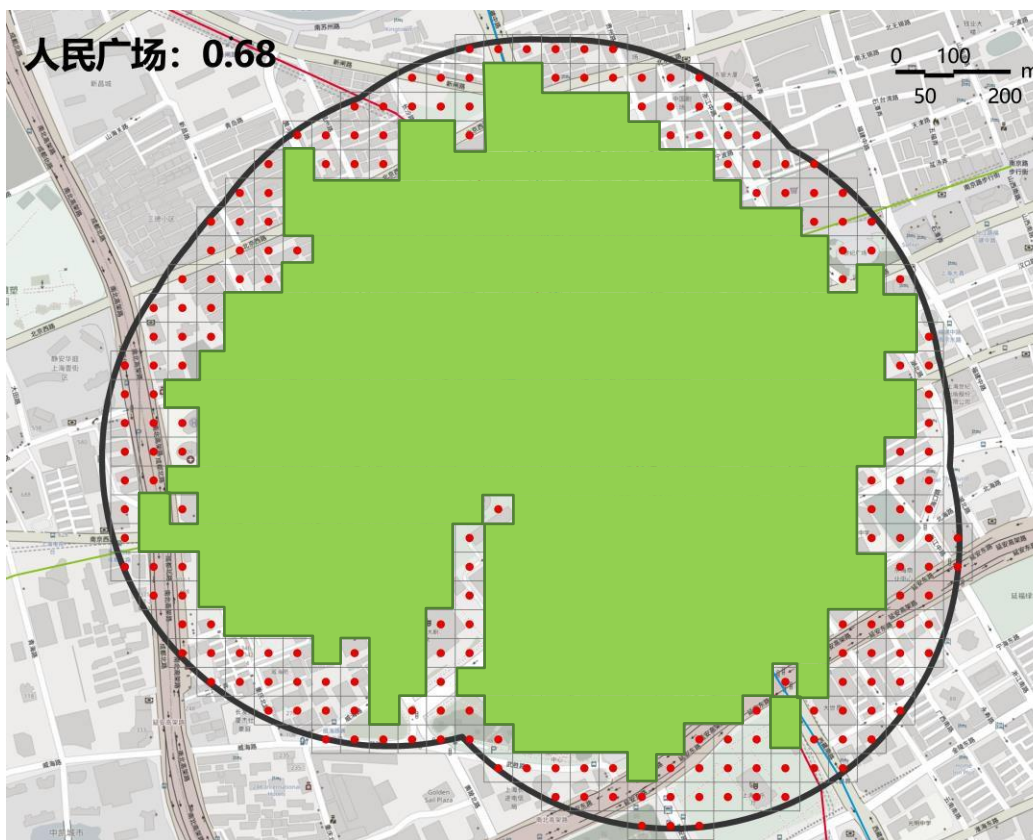
站点地区步行渗透区—有效服务区 Walk permeability

■ 影响步行覆盖率的因素

(1) **路网密度 (街区尺度)**。人民广场路网密度高达 $12.5\text{km}/\text{km}^2$ ，街区尺度约50-150m左右，而锦江乐园路网密度仅 $4.8\text{km}/\text{km}^2$ ，街区尺度约在200-300m，路网密度越高，步行覆盖率越大。步行覆盖率的大小与道路密度具有较弱的相关性，相关系数为**0.44**，说明除路网密度外，步行环境还与其他因素相关。

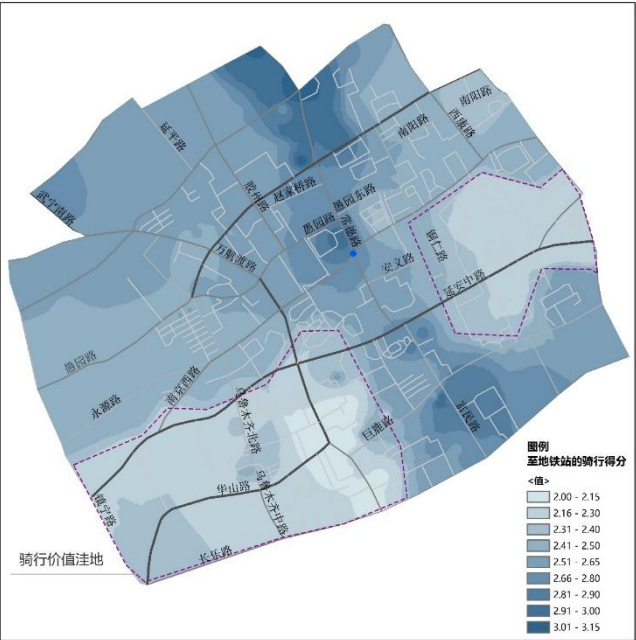
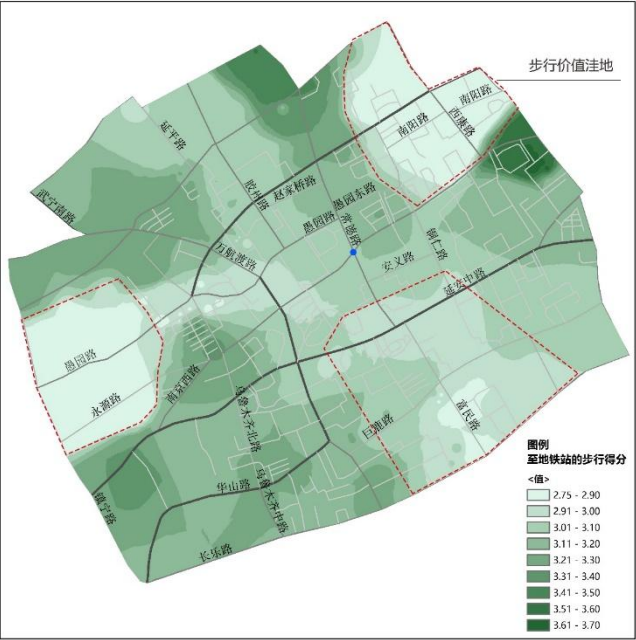
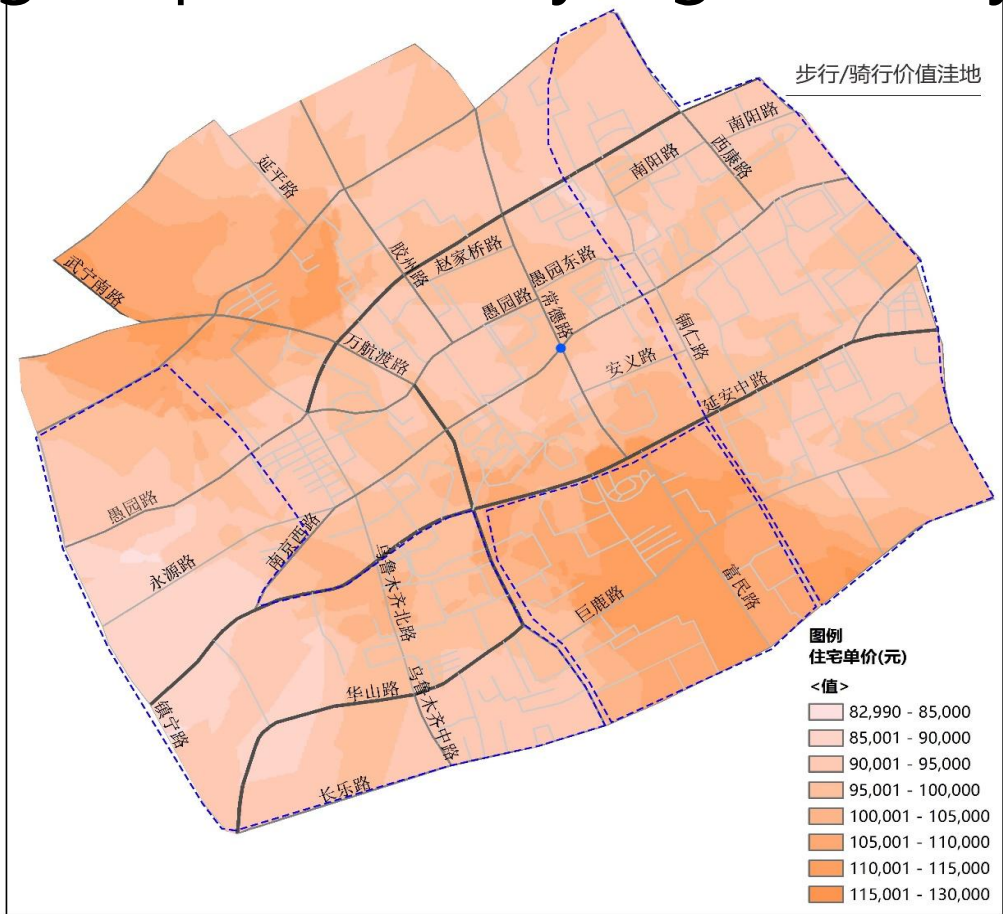
(2) **穿越性差的重要交通干道**。锦江乐园站位于中环路、沪闵高架路的交接处，同时有铁路线穿过，对步行的连续性产生阻隔，而人民广场周边主要以穿越性较好的支路为主；

(3) **自然条件**。河流、山体等自然条件也会影响步行的可达性。



人民广场与锦江乐园站500m步行覆盖率示意图

步行渗透区与站点地区地价--- 容积率，开发潜力，城市更新潜力 Higher permeability higher utility



改造重点

西北区域和东南区域的房价相对较高，单价在10万以上；西南区域和东北区域的房价相对较低，单价在9万以下。

房价较低的区域基本与步行/骑行得分的价值洼地相一致，仅有富民路两侧例外。

虽然富民路两侧区域至地铁站的步行得分较低，但其房价却较高。这是由于距离地铁站的距离这一变量在影响房价。



City governor

Vertical institution
construction infrastructure
Receive funds

Horizontal institution
District
Sub-District
Accept the products

建筑/城市设计
Archi-Design

轨道工程建设
Engineering

Swing-Style



城市规划/发展
Urban Development

Lost Space

“荡秋千式” 发展模式， 游魂式空间

愿景和价值导向

Social
Demographic
Data

Evidence,
Methodology

社会经济数据

城市规划/发展

Participation/Collaboration
No Solo Solution

多权益主体的参与

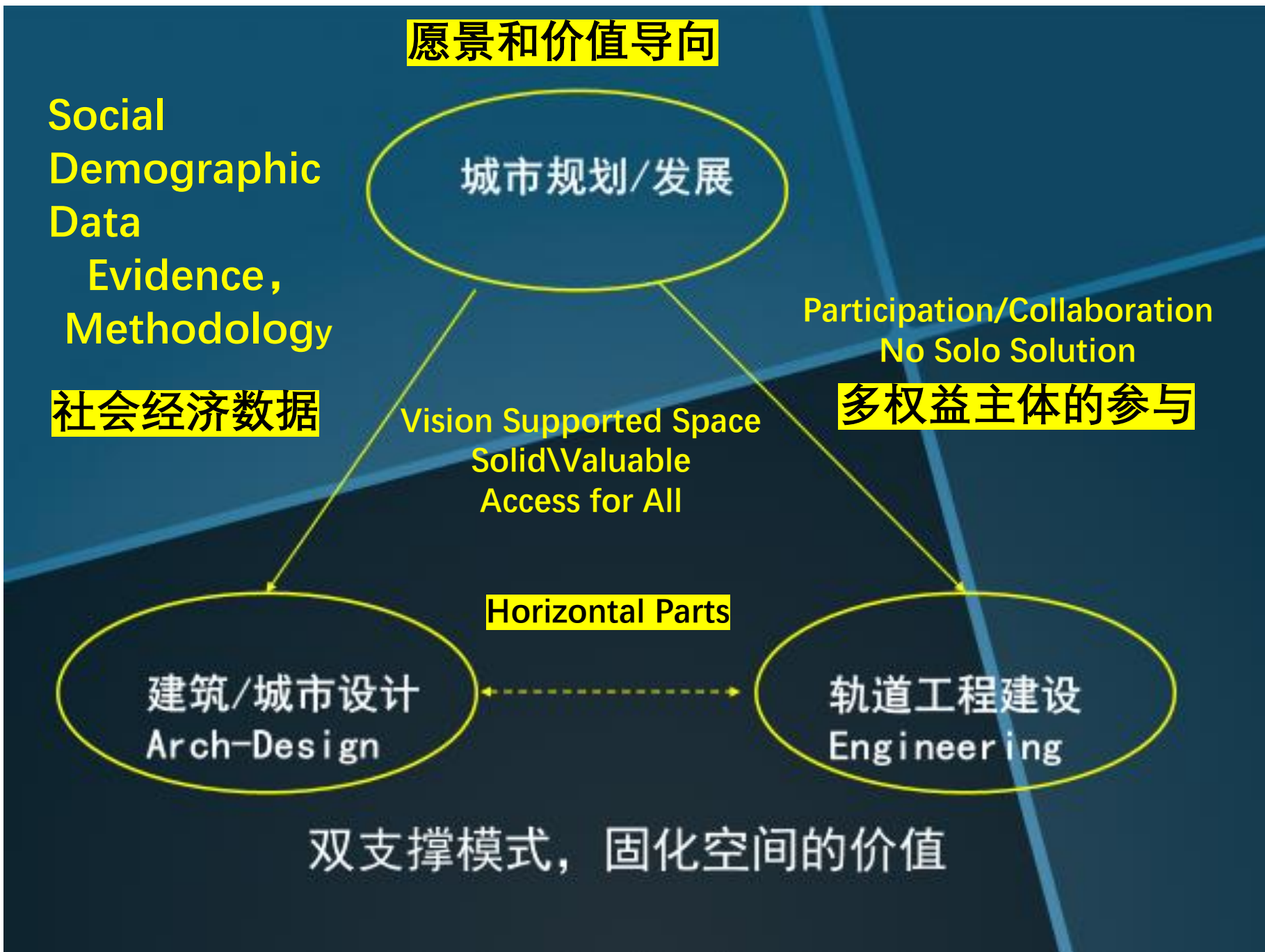
Vision Supported Space
Solid\Valuable
Access for All

Horizontal Parts

建筑/城市设计
Arch-Design

轨道工程建设
Engineering

双支撑模式，固化空间的价值



Thank You

Q&A